Project ELECTROJAN

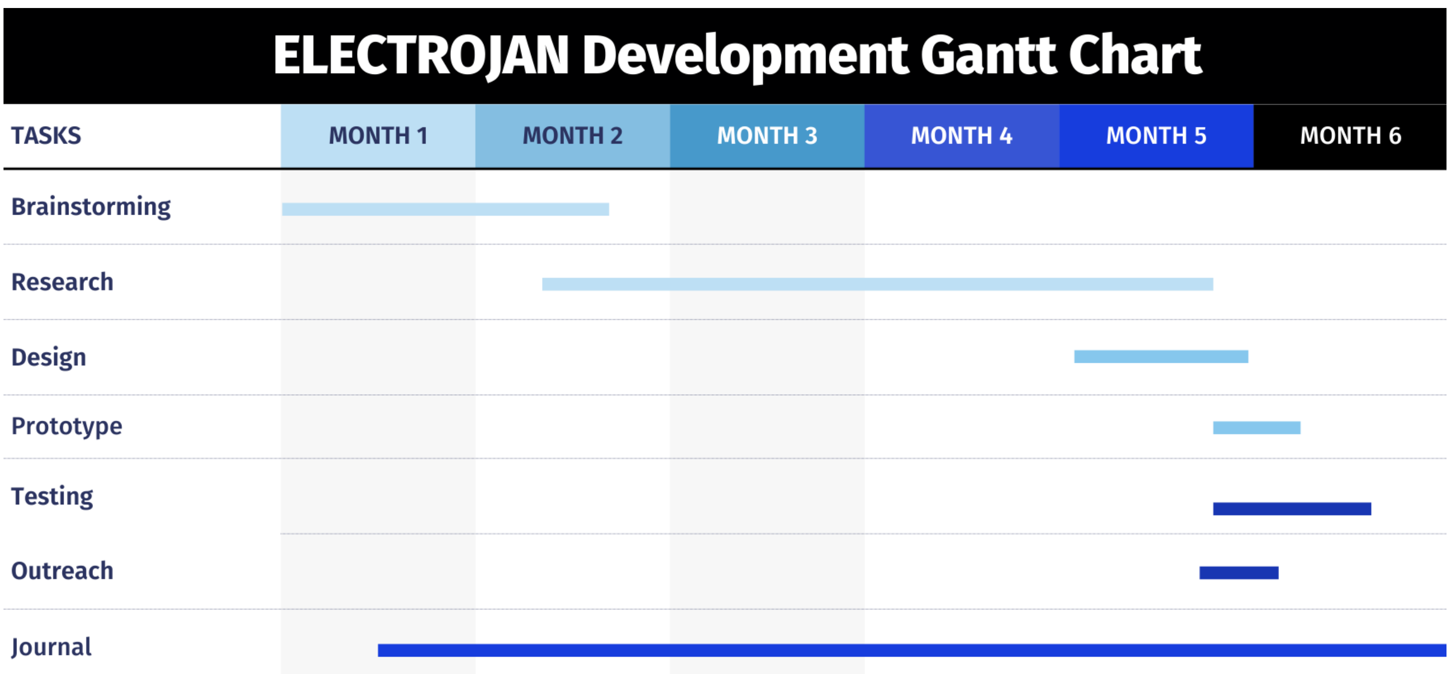
Protecting Every Connection

Coby Nguyen, Chase Nguyen, Nathan Viejo

SkillsUSA Engineering Technology Design Project

Table of Contents

Gantt Chart


ELECTROJAN Development Gantt Chart
TASKS
MONTH 1
MONTH 2
MONTH 3
MONTH 4
MONTH 5
MONTH 6
Brainstorming
Research
Design
Prototype
Testing
Outreach
Journal

Materials List

Capacitor 10 μF

Diode Rectifier 1N4007 x4

Resistor 100kΩ

Green 5mm LED

Metal prongs

Plastic – PLA Filament

Rubber – TPU filament

Spring

Wire

Engineering Journal

9/09/2024

Stage of Engineering Design Process: Forming

As we began researching concepts and problems to address for our Engineering Design Technology competition, we focused on safety concerns within the field of electrical engineering. Electricity is both essential and unpredictable, making it one of the most dangerous forces to handle without proper safeguards. To identify opportunities for innovation, we examined existing designs and standards, using them as inspiration for improvement.

Our primary focus is the safety of electrical plugs. American Type A and B plugs, commonly used in households, lack many of the safety features seen in plugs from other regions, such as the UK, where fuse-integrated plugs and protective shutters are standard. This disparity raises concerns about the adequacy of U.S. plug designs in preventing accidents like shocks or fires.

Additionally, industrial and construction environments present unique challenges, as their electrical plugs are subject to harsh conditions, including exposure to moisture, wear, and heavy usage. These issues demand a durable and reliable solution. Our project aims to improve the safety, usability, and standards of American Type A and B plugs while addressing the specialized needs of industrial and construction plugs. These were the main points for our project:

- Insulative cover for prongs when not plugged in, that push back when plugged in
    - Spark protection for gases
    - Lock in technology to prevent accidental unplugs and loose cables
    - LED indicator for grounding
    - Only activates when fully plugged in

Coby Nguyen

9/20/2024

Stage of Engineering Design Process: Forming

PROBLEM: It is very easy to accidentally shock yourself. Additionally, in construction, oil, and other labor environments, plugging in electronics can sometimes cause sparks which may interact with gases in the environment and result in an explosion.

RESEARCH:

https://www.pbs.org/wgbh/americanexperience/features/triangle-fire-deadliest-workplace-accidents/

https://onlinelibrary.wiley.com/doi/10.1155/2021/8451241

https://www.sciencedirect.com/science/article/abs/pii/S2214790X23000242

Retractable **Insulating Plug Prongs**

- **Concept**: Rather than a static insulating cover, design prongs that automatically retract into the plug housing when not inserted. This way, the prongs are only exposed when they're in the outlet, reducing the risk of accidental contact.
- **Key Features**: Spring-loaded mechanism, automatic retraction when unplugged, and a safety lock to prevent unintended retraction while in use. Locks in place to prevent accidental unplugs.

Dual-Layer Prong Insulation

- **Concept**: Use a dual layer insulative coating on plug prongs. The first layer is durable for general use, while the second is a retractable sleeve that only exposes the conductive surface when pushed into the outlet.
- **Key Features**: Shock-absorbing inner layer and heat-resistant outer layer to enhance durability and safety. The outer layer could include non-conductive materials that protect against ESD while handling.

Smart Outlet with ESD Protection

- **Concept**: Create an outlet with integrated ESD protection that neutralizes static discharge before a plug is fully inserted. This would be particularly useful in static-prone areas like workshops and labs.
- **Key Features**: Built-in ESD neutralization, LED indicators to show when the outlet is ready, and a detection system that only completes the circuit once the prongs are fully inserted.

Flexible Plug with Pressure-Activated Safety Prongs

- **Concept**: Design prongs that are insulated except for a small area at the tip, which only becomes conducive when pressure is applied (e.g., from insertion into an outlet).
- **Key Features**: Use pressure-sensitive materials or conductive polymers that only activate when pushed, ensuring safety if the plug is halfway inserted.

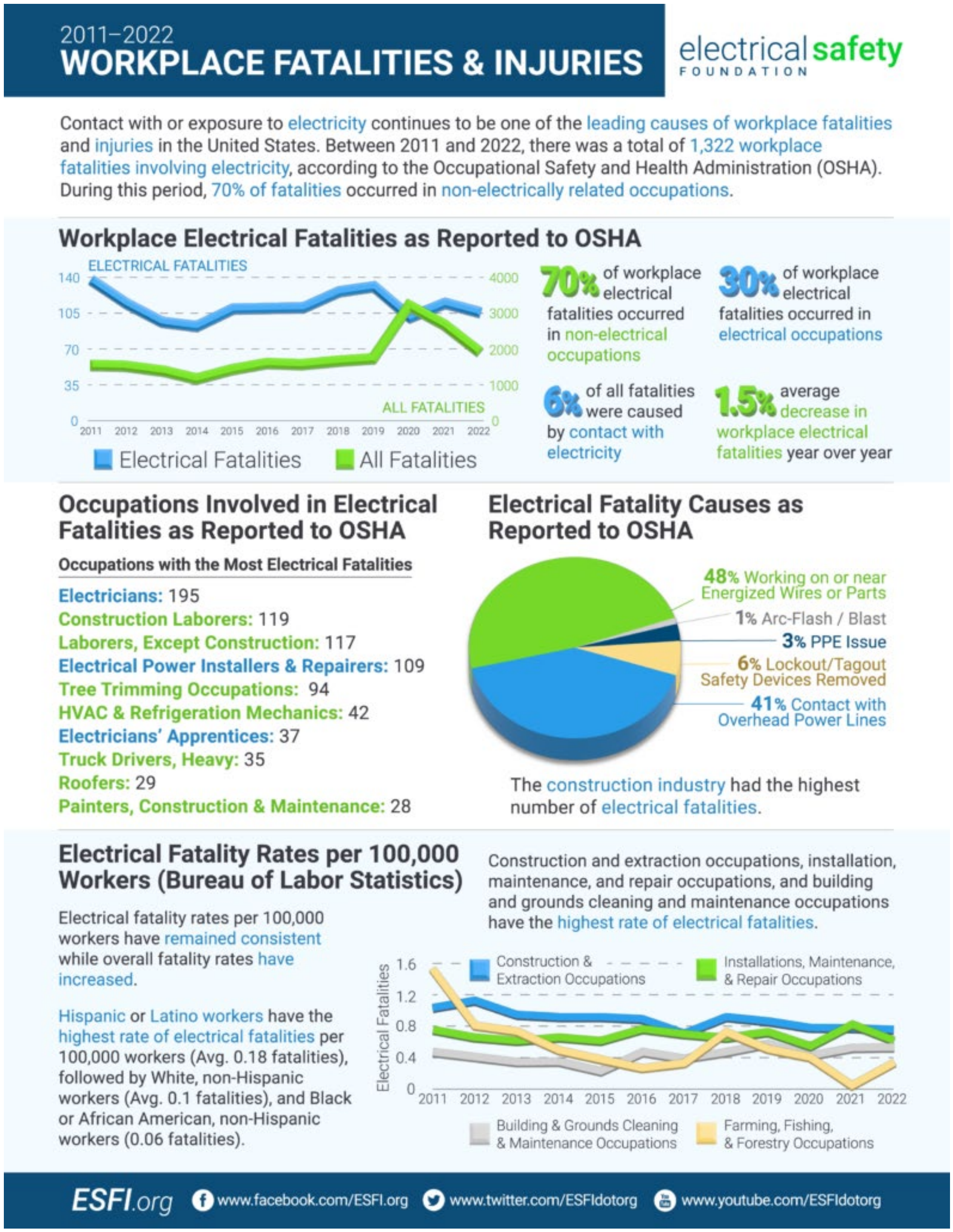
Anti-Arc Plug Design

- **Concept**: Develop a plug that has an arc suppression feature, reducing the risk of sparks or short circuits when partially plugged in or removed under load.
- **Key Features**: Integrate a small capacitor or suppressor circuit in the plug that neutralizes the arc upon plug removal, ideal for high-power appliances that might arc on disconnection.

Plug with Built-In Ground Testing

- **Concept**: Design a plug with a built-in indicator that verifies proper grounding before use, warning users of outlets with grounding issues before they connect sensitive devices.
- **Key Features**: Integrates a small LED or other indicator that lights up if grounding is detected, useful for reducing ESD and accidental grounding issues in faulty outlets.

Nathan Viejo



We decided to work on our electrical plug socket concept. We chose this concept because electrical safety impacts both the workplace environment and everyday life. Electricity can be unpredictable at times, especially for workers who are not electricians or familiar with how electricity works.

Coby Nguyen

10/29/2024

Stage of Engineering Design Process: Research

RESEARCH:

https://pmc.ncbi.nlm.nih.gov/articles/PMC8567394/

Through our research into electrical safety, we analyzed a case study on the piezoelectric effect in quartz-rich sandstone, where stress-induced electrical discharges led to hazardous outcomes such as methane ignition in mining environments. This phenomenon, caused by material properties and environmental stressors, highlights the importance of mitigating unforeseen electrical risks. These insights have significantly informed us of the development of our project concepts.

1. Mitigating Hazards Through Design

- **Source Insight:** Stress and deformation in quartz-rich sandstone resulted in unintended electrical discharges, creating safety risks.
- **Application:** This reinforces the necessity of proactive safety measures, such as **Retractable Insulating Plug Prongs** and **Dual-Layer Prong Insulation**, which minimize user contact with conductive elements and reduce the likelihood of electrical shocks.

2. Addressing Environmental Context

- **Source Insight:** Electrical discharges were pronounced in specific conditions, emphasizing the need for safety mechanisms in high-risk environments like workshops and labs.
- **Application:** Our **Smart Outlet with ESD Protection** neutralizes static discharge before the circuit is completed, offering a critical safety solution in static-prone settings.

3. Controlled Activation of Conductive Elements

- **Source Insight:** Uncontrolled energy surges in the material highlighted the dangers of latent electrical activity.
- **Application:** Concepts like **Flexible Plug with Pressure-Activated Safety Prongs** and **Anti-Arc Plug Design** ensure conductive elements only activate when fully inserted into an outlet, preventing hazards during partial use or improper handling.

4. Enhancing Detection and Feedback Systems

- **Source Insight:** The unpredictable nature of piezoelectric discharges stresses the importance of early detection and user feedback.
- **Application:** The **Plug with Built-In Ground Testing** provides real-time feedback on grounding issues, helping users identify and address potential electrical risks before operation. Similarly, **Smart Outlets** incorporate LED indicators to confirm readiness and safety status.

Impact on Our Project Concepts

This research underscores the importance of anticipating and mitigating risks associated with electrical systems. By integrating proactive safety mechanisms into our designs, we aim to address common challenges such as accidental shocks, static discharges, and arc flashes. Each concept emphasizes safety, innovation, and practicality:

1. **Retractable Insulating Plug Prongs** ensure prongs are only exposed during use.
2. **Dual-Layer Prong Insulation** adds extra protection with durable coatings and retractable sleeves.
3. **Smart Outlets with ESD Protection** neutralize static discharge and enhance user feedback with LEDs.
4. **Pressure-Activated Safety Prongs** prevent electrical conduction until full insertion and activation.
5. **Anti-Arc Plug Design** reduces sparks and short circuits during disconnecting.
6. **Built-In Ground Testing** alerts users to grounding issues in faulty outlets.

Nathan Viejo

11/18/2024

Stage of Engineering Design Process: Research

https://www.cdc.gov/niosh/mining/topics/fires-explosions.html

https://www.cdc.gov/niosh/mining/topics/electrical-injuries.html

Today, we examined the hazards associated with mining fires and explosions, focusing on the risks posed by methane buildup, ventilation issues, and inadequate rock dusting. We analyzed the methods used to mitigate these risks and how modern technologies are being implemented to improve safety in underground mining operations.

The NIOSH Mining report highlights the significant dangers of electrical injuries in mining, emphasizing that electrical accidents are a leading cause of fatalities in the industry. According to data from the U.S. Bureau of Labor Statistics, the fatality rate for electrical accidents in mining is 8 to 12 times higher than in other U.S. industries. Additionally, electrical injuries in mining are disproportionately deadly, with one fatality occurring for every 22 electrical-related injuries, compared to one fatality for every 203 injuries from other mining-related hazards. This information underscores the critical need for improved electrical safety measures in high-risk environments, making it highly relevant to our research and project.

Similarly, research from other sources has emphasized the importance of electrical safety in various industries, particularly in high-risk environments such as construction and manufacturing. Electrical hazards, including improper wiring, arc flashes, and inadequate protective equipment, remain persistent issues that contribute to severe injuries and fatalities. The need for better safety protocols, enhanced protective gear, and improved electrical system designs is evident in these reports, further reinforcing the objectives of our Engineering Technology Design project.

Our project focuses on enhancing electrical safety through innovative solutions, and the findings from NIOSH and other sources provide a strong foundation for our work. They outline key areas where safety improvements are needed, such as power system design, engineering controls, work organization and procedures, protective equipment, and human factors. By addressing these concerns, we can develop solutions that mitigate risks associated with electrical exposure, particularly in hazardous industries like mining and construction.

The reports also emphasize ongoing efforts to prevent electrical injuries, such as improving training programs, enhancing battery safety, and designing safer switching equipment and cables. This aligns with our approach of integrating advanced safety mechanisms into electrical systems. For example, our project aims to incorporate LED indicators for outlets and multi-wire plugs to provide real-time feedback on connection status, reducing the risk of accidental shocks. Additionally, our focus on insulative covers for plug prongs directly contributes to minimizing metal exposure, which is a known hazard in industrial settings. Furthermore, redesigning outlets to be more durable and resistant to environmental hazards such as moisture and wear enhances overall safety.

Understanding the severe consequences of electrical accidents across multiple industries reinforces the importance of proactive safety measures. By incorporating industry research into our project, we ensure that our solutions are grounded in real-world data and address critical gaps in electrical safety. This journal entry serves as a reflection on how our research is supported by industry findings, guiding us toward developing effective, practical, and life-saving innovations in electrical safety.

Nathan Viejo

11/20/2024

Stage of Engineering Design Process: Research

Today, we collaborated with our fellow seniors, Armando and Victorino, to brainstorm ideas for improving electrical safety. One of the key issues we discussed was the questionable reliability of electricity flow in certain settings, particularly manufacturing and industrial environments. These areas frequently face challenges with outlet wear and tears, exposure to moisture, and the complexity of using specialized plugs for high-voltage equipment.

Key Ideas Discussed:

1. **LED Indicators for Connection Status**
    - Outlets are equipped with LEDs to show whether they are connected to a power source and if current is flowing. This provides a quick and easy way to identify functional outlets and connections.
2. **Industrial-Grade Outlet Redesign**
    - Outlets redesigned for enhanced durability to withstand frequent use, exposure to wet conditions, and the demands of industrial and construction environments.
    - Include options for different plug standards based on specific applications:
        - Home use (double prong, standard voltage).
        - Construction use (triple prong, high voltage).
        - Specialized industrial plugs for heavy machinery.
3. **Phone-Connected Outlet System**
    - A smart system that connects to a phone app, allowing users to identify which fuse switch controls a specific outlet. This would simplify maintenance and troubleshooting.
4. **LED Indicators for Multi-Wire Plugs**
    - For plugs with multiple wires, integrate LEDs to show which wires are actively connected. This ensures accurate and safe connections, reducing risks of error or shock.
5. **Simplifying Construction Site Wiring**

- Address the complexity of construction site wiring, where workers often must wrap wires around equipment or use non-standard plug styles. A standardized, high-durability outlet system could significantly reduce complications.

Manufacturing Considerations:

We also considered the manufacturing process for these ideas. For example, outlets could be designed using high-grade, moisture-resistant plastics and durable internal components to minimize failure rates. Production could involve injection molding for outer casings and precision assembly for electronic components like LEDs and sensors. By creating modular designs, manufacturers could offer interchangeable components for specific applications, further extending the lifespan of the outlets and reducing waste.

Outlets burning out or malfunctioning due to wear and tear or exposure to wet conditions is a common issue, particularly in demanding environments. Our innovations aim to tackle these problems while introducing user-friendly features like visual indicators and smart connectivity.

Chase Nguyen

11/26/2024

Stage of Engineering Design Process: Research

Today we researched how American plugs and outlets work, and how they compare to standards across the world. This allows us to implement their design philosophy into our standards and ensure we can use the best methods.

America uses Type A and Type B plugs. The main consideration and focus of these plug designs are to be cost effective

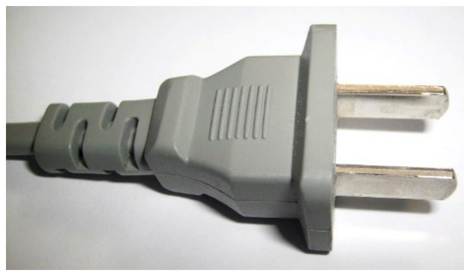


Figure 1. Type A ungrounded plug

Figure 2. Type B grounded plug

The cover for both plugs will be relatively the same since the ground prong does not need a cover since they do not pose a direct shock hazard. Type A and B plugs are known to be dangerous due to the fact they are not insulated or recessed into the wall, meaning power still flows when the plug is halfway pulled out. This means if a plug is not fully secured, it is easy for a person or piece of metal to touch the pins.

Figure 3. Type B plug halfway inserted but still retains power

Chase Nguyen

12/04/2025

Stage of Engineering Design Process: Design

Using the planning stage of the Engineering Design Process, we collaborated as a group to all plan and understand how our project will function based upon a hand-written sketch. By brainstorming the workings of Electrojan as we made the sketch, we can see how our project demonstrates the benefit of preventing electrical shock with the use of Electrojan's prong-covering mechanism.

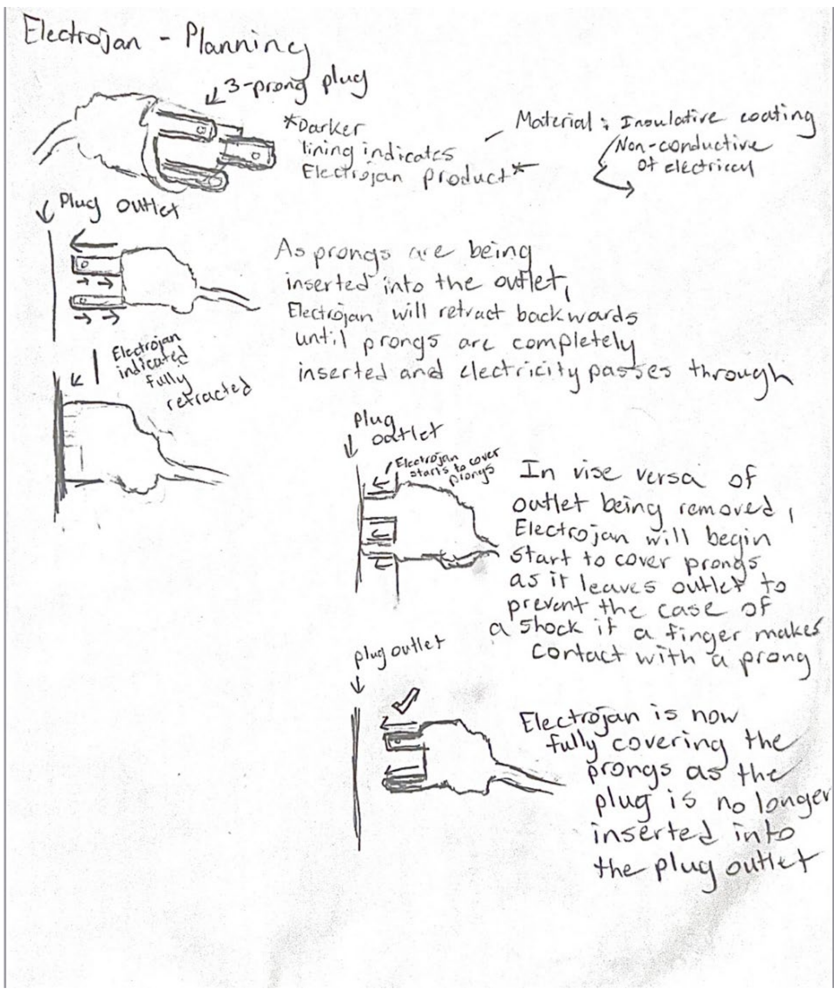


Figure 1. Sketch of Electrojan

Nathan Viejo

12/15/2025

Stage of Engineering Design Process: Research

To optimize the features of our plug standards, we investigated the workings of different plug designs across the world. The United States currently uses Type A and B plugs, which we researched earlier. The main problems with these plugs are their lack of secure insulation, insufficient grounding, and the fact that they do not fully recess into the wall, leaving the prongs exposed and increasing the risk of shock or accidental disconnection. By studying international plug designs, we can incorporate safety and usability improvements into our design.

Type C

Used in the majority of Europe and one of the most common plug types worldwide alongside Type A and B. This is a two-pronged, ungrounded, and unpolarized plug. Its design features a rounded plastic or rubber base, which makes it easier to grip and reduces wear on the cable. Type C plugs are compatible with recessed outlets, enhancing their safety by preventing accidental contact with live prongs.

Type F

Popular in Germany, Russia, and many European countries, Type F plugs are grounded with two metal clips on the side that provide a secure earth connection. The prongs are slightly thicker and longer than Type C, ensuring better contact. They also feature insulated pins to prevent accidental shocks when plugging in or unplugging. This design prioritizes grounding and security, making it safer for high-wattage devices.

Type G

Primarily used in the United Kingdom, Ireland, and several Commonwealth nations, Type G plugs are renowned for their safety features. They are rectangular, grounded, and include a built-in fuse in the plug itself to prevent overcurrent damage, and are replaceable. The prongs are long and thick, with insulated bases to prevent contact with live pins. Additionally, outlets for Type G plugs have safety shutters that only open when all three prongs are inserted simultaneously. This design emphasizes durability, grounding, and child safety.

Type I

Used mainly in Australia, New Zealand, Argentina, and China, Type I plugs have a unique triangular prong layout. They feature insulated live and neutral pins, reducing the risk of shock when the plug is not fully inserted. The plugs are polarized and grounded, and the prongs are slightly flat to ensure a more secure connection. These plugs often include strain relief designs at the cable entry point, increasing durability under frequent use.

Type N

Commonly used in Brazil and South Africa, Type N plugs are grounded and feature three round pins arranged in a triangular pattern. They are specifically designed to work with recessed outlets, preventing accidental contact with live parts. The prongs are sized to match the outlet precisely, ensuring a snug fit that reduces wear and disconnection issues.

Key Observations

1. **Safety through insulation:** Many plugs, such as Type I and F, incorporate insulated prongs to prevent accidental shocks during use.
2. **Grounding and polarization:** Grounded plugs like Type G and F ensure safer connections for high-wattage devices, while polarized designs like Type I prevent incorrect wiring. We plan to also use an LED to indicate a secure connection.
3. **Recessed outlets:** Plugs designed for recessed outlets, such as Type C and N, offer better protection by minimizing prong exposure.
4. **Lock-in mechanisms:** Some plugs, like Type F, include holes or clips on the prongs to ensure they lock securely into the outlet, preventing accidental disconnection. Some type A and B clubs have larger ends of the prong too.
5. **Fuses and shutters:** Type G plugs stand out for their built-in fuses and safety shutters, providing additional protection against electrical faults and ensuring child safety.

These design nuances, including prong insulation, secure locking mechanisms, and recessed compatibility, highlight how international standards have prioritized safety and durability. By integrating these features into our improved plug design, we aim to address the shortcomings of Type A and B plugs while providing a safer and more reliable solution.

Chase Nguyen

1/06/2025

Stage of Engineering Design Process: Design

Today we did another sketch of our plug with updated features from our past research.

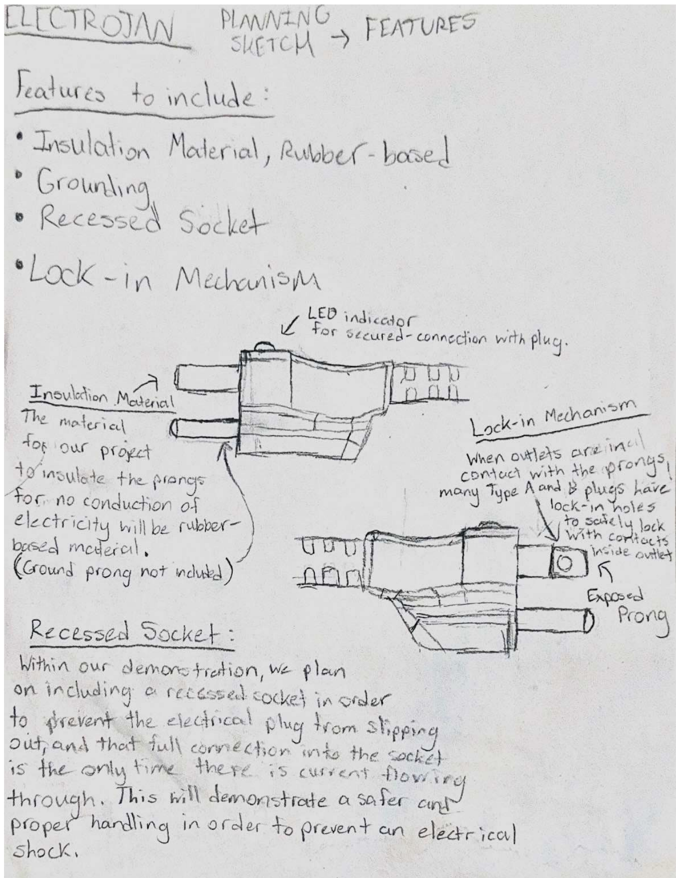


Figure 1. Second sketch of our plug

The main features of our outlet are:

- Retractable prong covers
- Insulative material on the prongs
- Holes at the end allowing for locking into outlets
- LED indicator for secure connections
- Longer ground prong

Nathan Viejo

1/06/2025

Stage of Engineering Design Process: Design

We developed a mockup of our design in SolidWorks to see what we would need to change into a Type B plug for it to have the retractable cover.

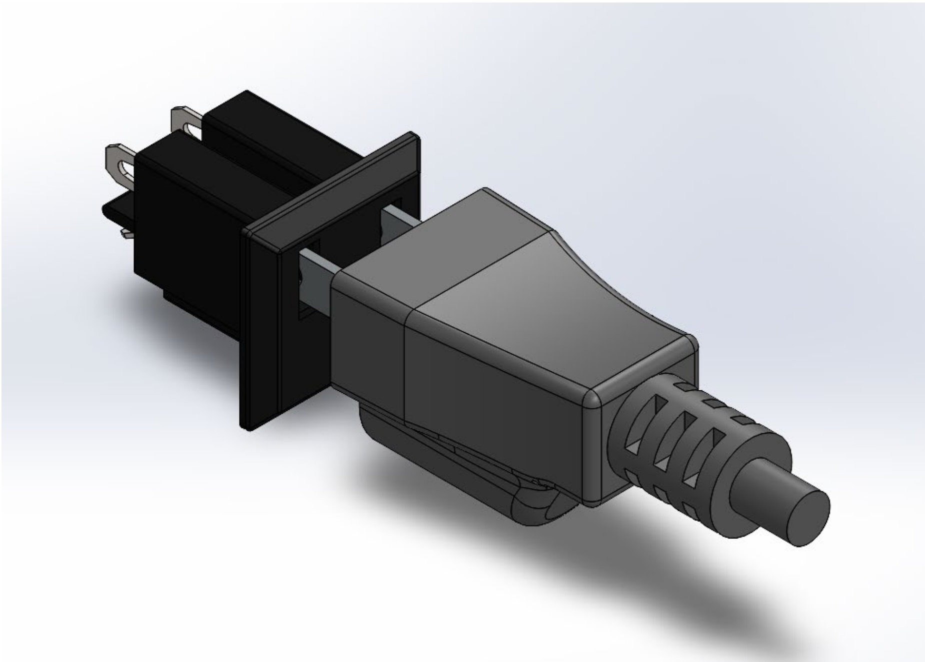


Figure 1. Isometric view of plug while plugging in

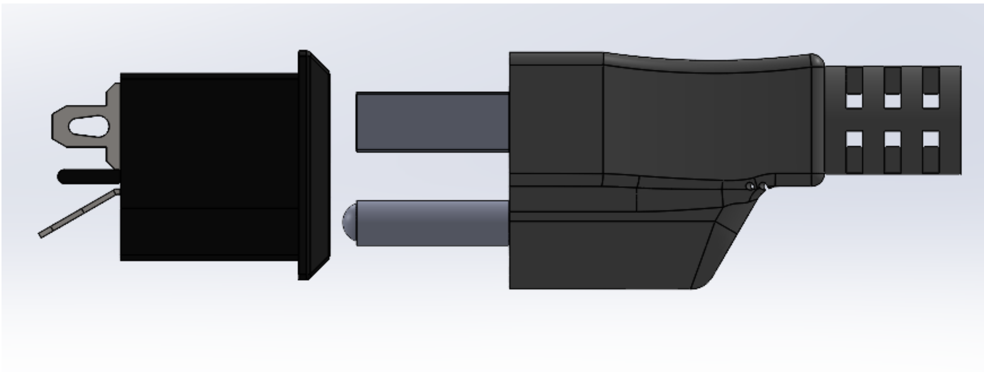


Figure 2. Side view of plug unplugged

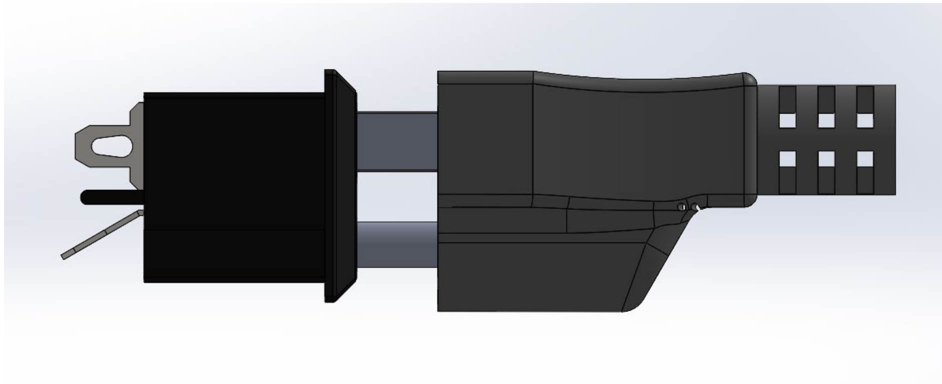


Figure 3. Side view of plug while plugging in

Coby Nguyen

1/8/2025

Stage of Engineering Design Process: Research

Today, upon further research, we spent our time watching, via YouTube, on the workings and components of an electrical outlet. This research is an important step to our project as we need to know how electricity is directed and current flows, which by potentially using an electrical outlet to present the function of the prong covering safety, we can bring it fourth into our demonstration.

The UK Type G plug has many additional features for safety that we found interesting. Not only is the ground prong longer than the others, but the outlets also themselves are locked from the inside until the longer ground prong is inserted. There are shutters to prevent anything from entering from the live side, until the device is properly grounded. Their plugs also have the cord coming from the bottom, rather than the back of the plug. This is to discourage people from pulling or yanking the wire out of the outlet, which is a major concern and problem we have seen people do, especially in the US.

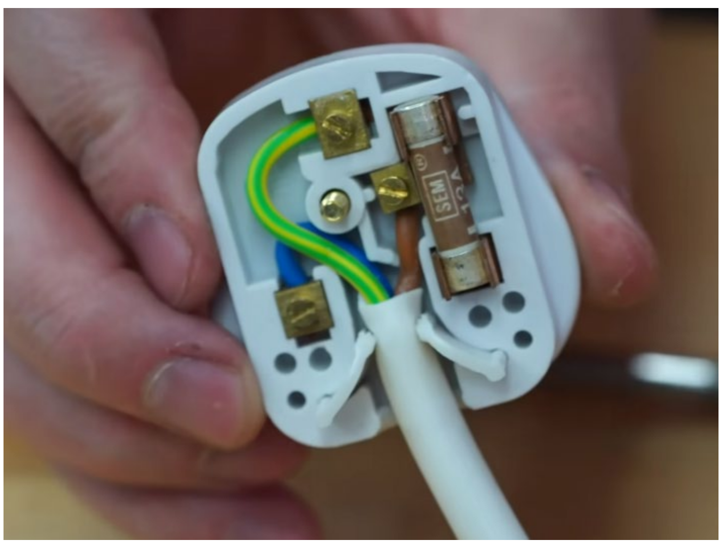


Figure 1. The inside of a Type G UK plug.

The inside of a Type G plug is also very interesting. The backing case of the plug is removable for easy replacement of the fuses and wiring in case of any issues. In the case that the entire cord is somehow pulled out, the first to come out are the live wires, meaning that the device will be grounded until all the wires are pulled out. There is also built in strain relief with two smaller fins on the cable, which can also be replaced.

BecomingAnElectrician. *How Electrical Plugs Work (Wire an Outlet)*. YouTube, 20 Mar. 2023, https://youtu.be/EOep646avQE?si=N0InSEYCdBg3CwsN.

LRN2DIY. *British Plugs and Outlets Are On Another Level*. YouTube, 09 Oct. 2022, https://youtu.be/139Q61ty4C0?si=_IJvTGP2HoLjXUOV

Sam's Tech. *American vs European Power Sockets: What's The Difference?*. YouTube, 07 Apr. 2023, https://youtu.be/kPMd8NWYc6o?si=waeEr4YXDniuuPWn.

The Engineering Mindset. *How Do Power Outlets Work?*. YouTube, 18 Oct. 2022, https://youtu.be/Tglht_V4fzo.

Tom Scott. *British Plugs Are Better Than All Other Plugs, And Here's Why*. YouTube, 07 Jul. 2014, https://youtu.be/UEfP1OKKz_Q?si=93sx4a0HIJns7DUL

Nathan Viejo

1/11/2025

Stage of Engineering Design Process: Research

To optimize the design of our prong covers, we investigated the inside of a cable. This allowed us to visualize how our covering system will be implemented into the current Type A and B standards. From our research, we learned that the Type B plugs include the hot wire, which is what carries the flow of the current, the neutral wire flows the current to the source that's connected with the plug, and the ground wiring is for protection of electrical shock by bringing escaped electricity strictly down to Earth's ground when there's a faulty occurrence. Learning about these features on the Type B plug is important to our project, as we got to learn about the grounding feature that's already built-in as a source of safety

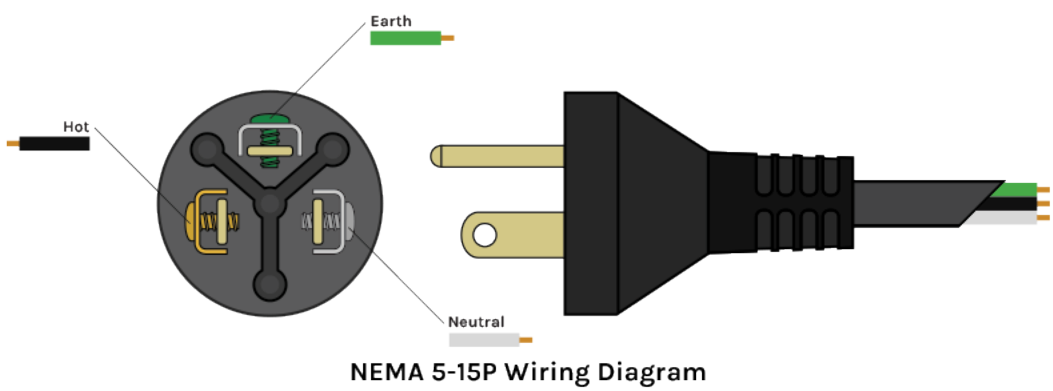


Figure 1. Wiring diagram of a Type B cable.

As for the Type A plug in comparison to the Type B plug, the Type A plug only includes the hot and neutral wire which is what lacks the safety feature of an implemented ground wiring with the third prong, causing higher risk of a potential electrical shock. This will help us to discuss features that we can implement into our project that is universal to both Type A and B plugs for the benefit of their use in our everyday lives.

Nathan Viejo

1/16/2025

Stage of Engineering Design Process: Design

Today, we made considerable progress in refining our design for the electrical socket with a retractable cover. We decided to remove the cover for the ground prong because the pin provides no direct shock risk and safely discharges electricity.

By integrating the insights gained from our SolidWorks mockup and the detailed study of electrical outlet components, we managed to identify key areas where our design could be optimized for better functionality and safety. We plan to add further features from the other types we researched previously. In our next design we intend to add holes and insulation on the prongs. We also want to research methods of preventing accidental disconnection and wear and tear through those disconnections.

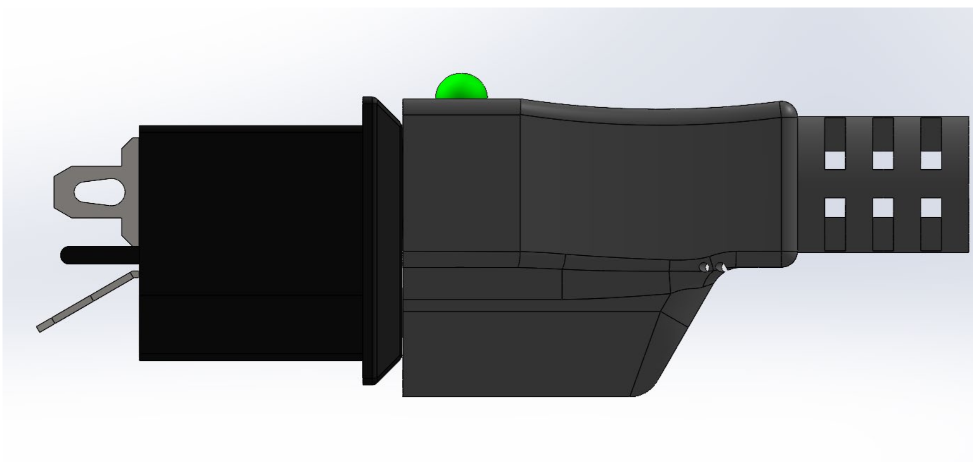


Figure 1. Updated design of Type B plug. It has a green LED that lights up when there is a secure connection.

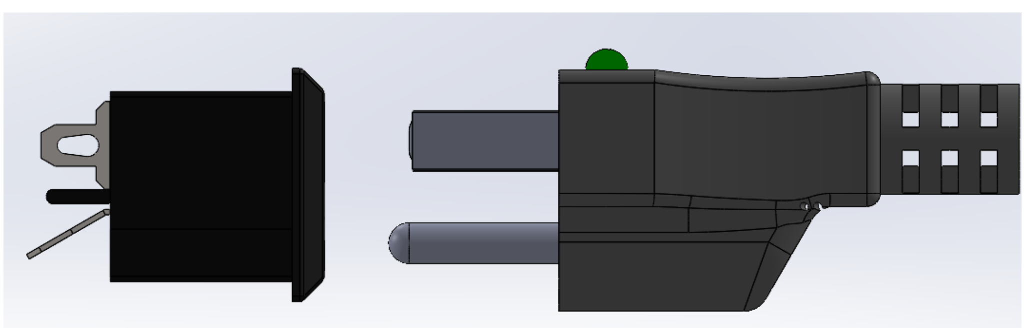


Figure 2. Updated design of Type B plug. The green LED does not light up when there is no connection or when the connection is not secure

Coby Nguyen

1/19/2025

Stage of Engineering Design Process: Design

Today, we focused on incorporating additional visible features into the design of our plug. These updates aim to enhance safety and functionality. The key improvements include:

1. **Polarization:**
    - Our plug is now polarized, meaning one prong is wider than the other. This design reduces the risk of electrical hazards, such as sparks or fires, in the event of a fault in the cord, outlet, or connected device.
2. **Holes at the Ends of the Prongs:**
    - We added small holes at the ends of the prongs, allowing them to be more securely fastened in outlets that support this feature. This ensures a stronger physical connection and helps to minimize the chance of accidental unplugging. We take this design from the Type B prongs where some of their designs have these holes, but not all do. We decided to add this to part of the updated standards.

These changes align with our goals of improving electrical safety while enhancing the overall user experience. We will continue refining the design to explore additional features that address both safety and durability.

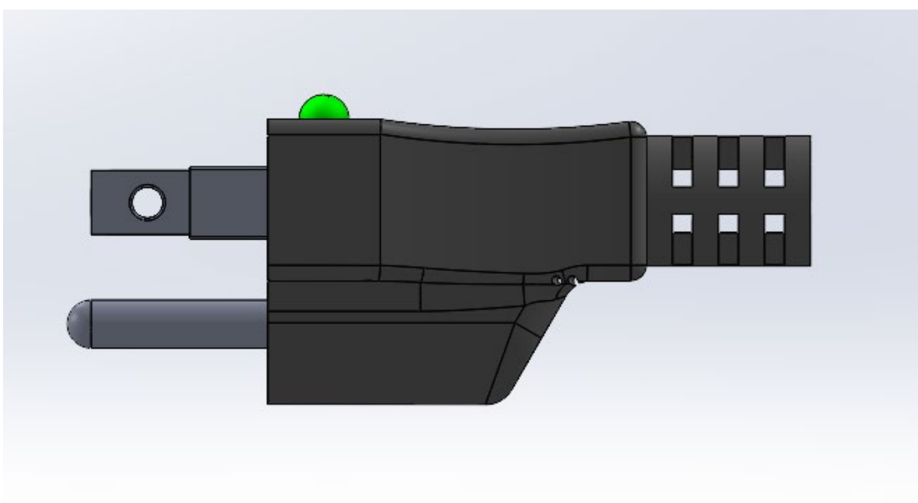


Figure 1. Side profile of the plug

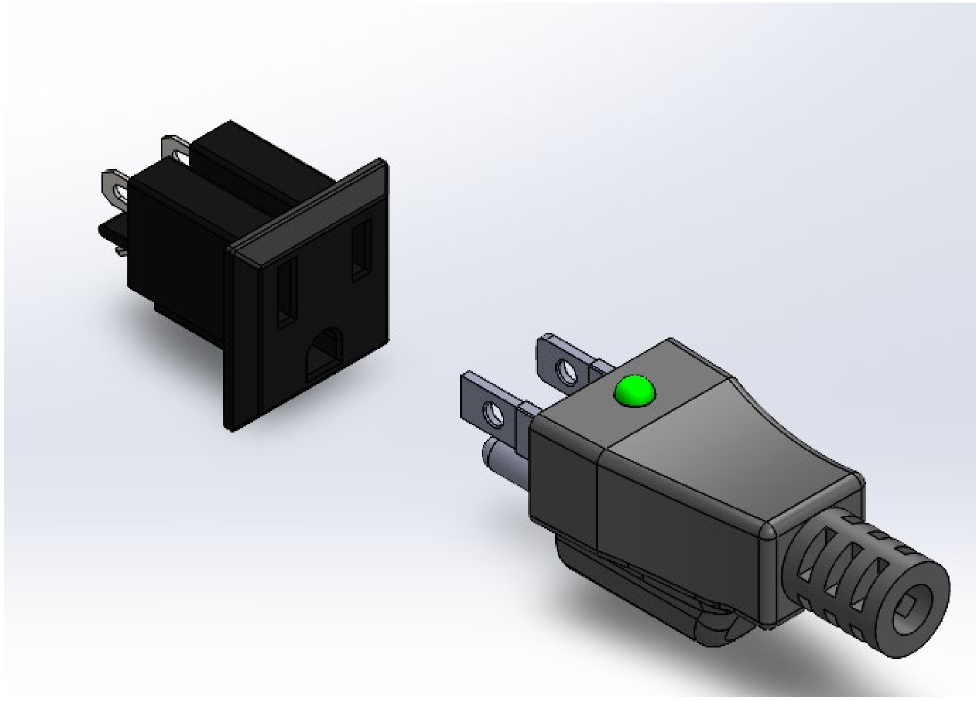


Figure 2. Isometric view of the plug and outlet

Figure 3. Exploded view of the plug assembly (without wiring)

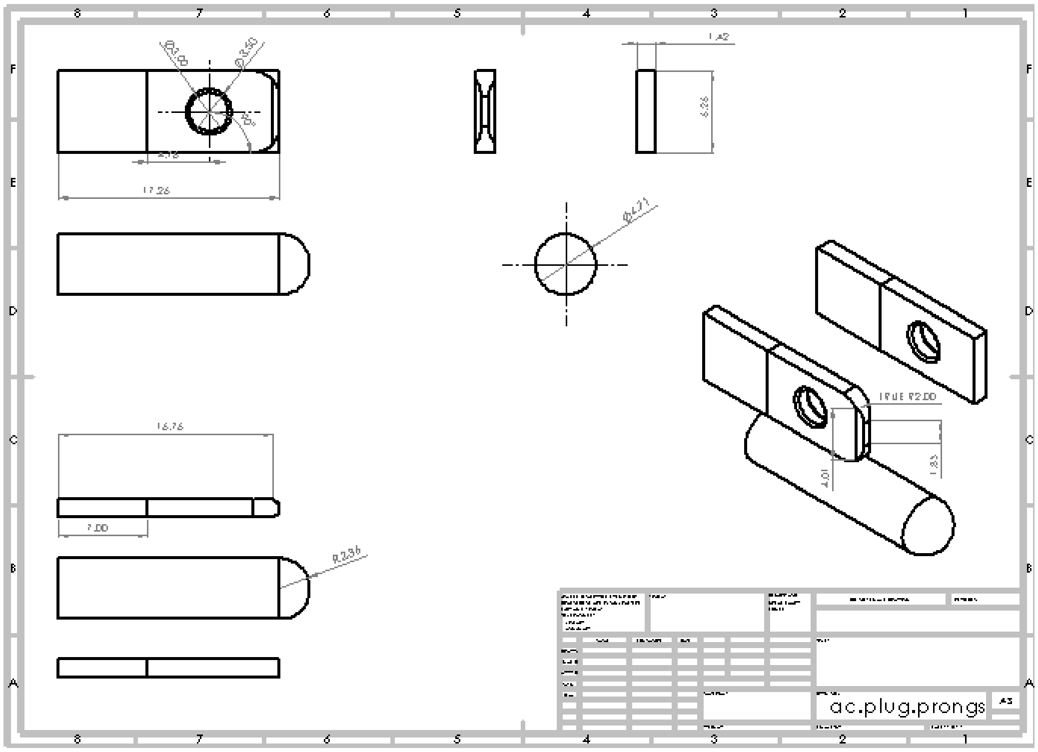


Figure 4. Technical drawing of prongs

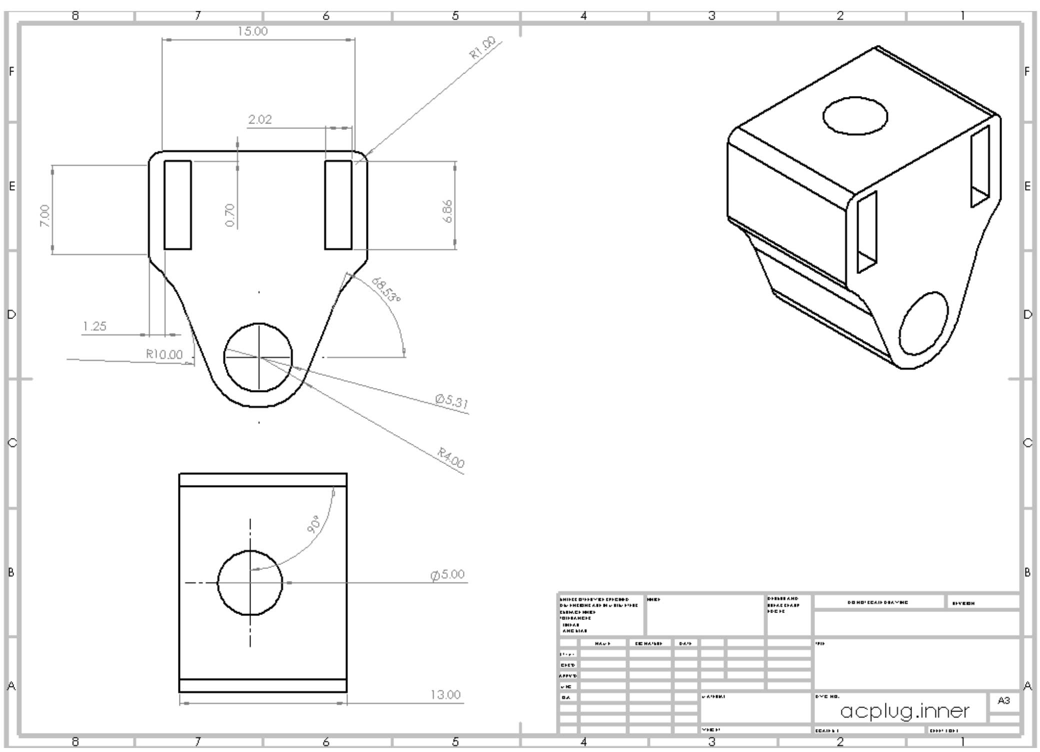


Figure 5. Technical drawing of inner outlet casing

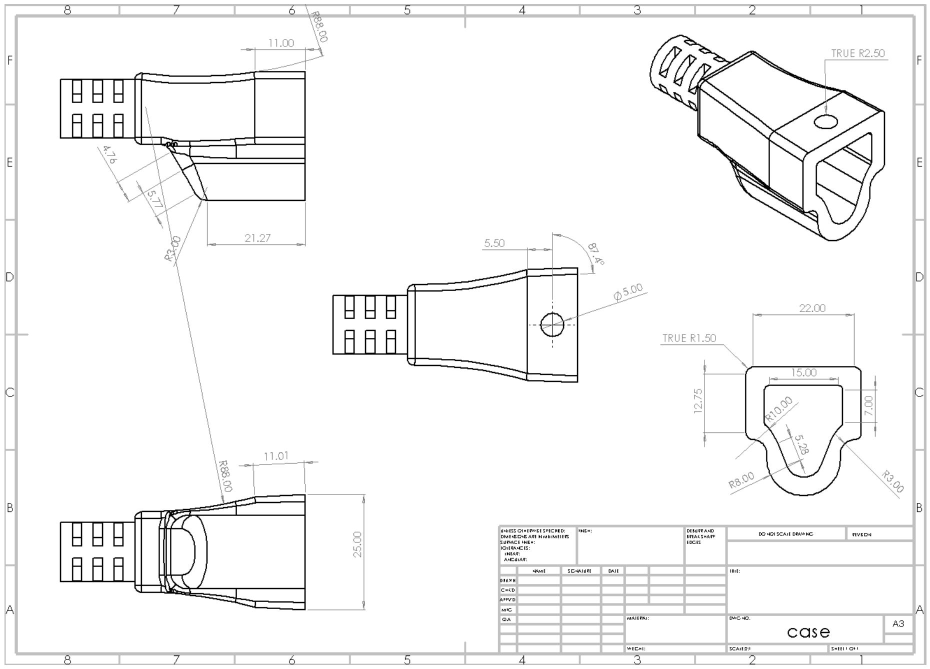


Figure 6. Technical drawing of outer outlet casing

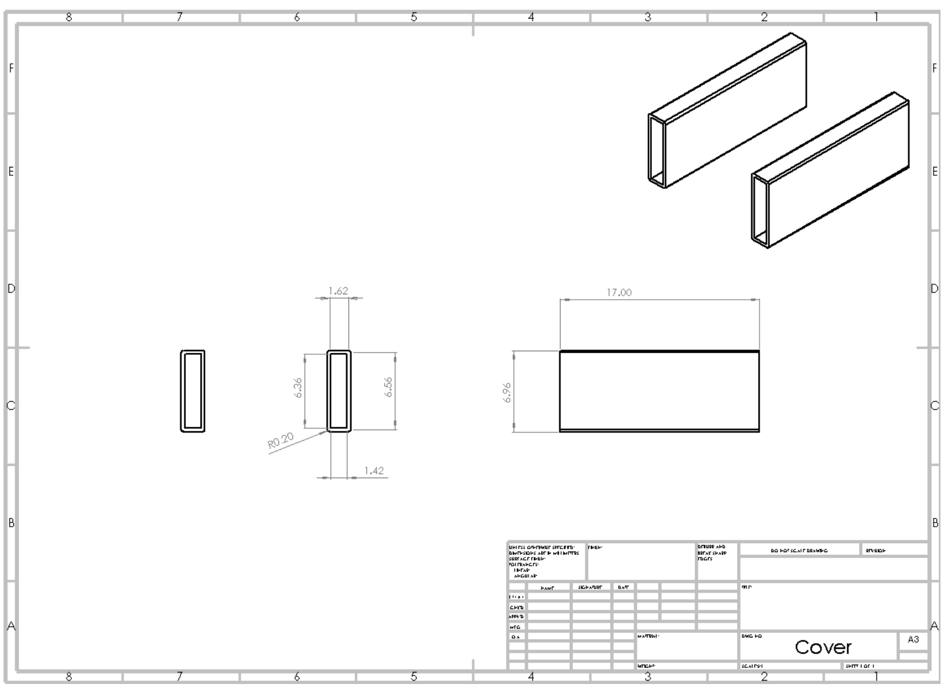


Figure 7. Technical drawing of the prong covers

Coby Nguyen

1/22/2025

Stage of Engineering Design Process: Research

We brainstormed more features that we would potentially implement into our plug design. The main concern is for potential sparks, especially with heavy duty and industrial plugs. There are two solutions we had:

Capacitors in Plug Design

- Capacitors store and release energy to smooth voltage fluctuations, suppress noise, or prevent arcing.
- Applications:
  - Arc suppression to minimize sparking during plug insertion/removal.
  - Stabilizing power delivery to reduce wear on connected devices.
- Advantages:
  - Improve safety by reducing electrical arcing.
  - Protects appliances from voltage spikes or surges.
- Challenges:
  - Precision engineering is required to match plug power ratings.
  - Limited lifespan—failure over time may reduce reliability.
  - Higher cost and complexity than fuse designs.
- Viability: Capacitors are more suitable for specialized applications, such as industrial plugs or smart plugs with advanced features, but less practical for standard household plugs.

Potential Applications for Project Design

1. Fuse Plugs:
   - Use a small, replaceable fuse to protect against overloads or short circuits.
   - Pair with LED indicators to display fuse status (e.g., green = intact, red = blown).
   - Ideal for industrial grade plugs or environments where safety is paramount.
   - Used in Type G plugs
2. Capacitor Plugs:
   - Integrate a small capacitor to suppress arcs and stabilize voltage.
   - Suitable for high-power appliances or plugs exposed to fluctuating power sources.
   - Compatible with anti-arc plug designs for construction or manufacturing use.

Integrating a fuse into plug designs is simpler, cost-effective, and practical for overload protection, making it an excellent choice for industrial or household applications. Capacitors, while more complex, offer advanced features like arc suppression and voltage stabilization, making them valuable for specialized or smart plugs.

The primary application of integrating fuses or capacitors into plug designs would be for industrial environments. As discussed by our seniors, industrial settings often require enhanced safety measures to manage high power demands and rugged conditions, and they would like if their plugs had a proper indicator for whether they are functioning or not.

Due to these comparisons, we have decided to implement a fuse in our plug design because of its prevalence in the United Kingdom in Type G plugs and its cost compared to the amount of safety provides. This allows us to ensure maximum safety while keeping the plug easy to manufacture. The added benefit of using a fuse is that if the plug has a fault, you can easily open the plug to replace the fuse with minimal effort. UK plugs use 13A fuses, so our fuse will likely be smaller since we se 120v instead of 220v. For now, our plug will use a 10A fuse.

With these current ideas, here are the key features of our current plug design:

- Retractable prong covers
- Insulative material on the prongs
- Holes at the end allowing for locking into outlets
- LED indicator for secure connections
- Longer ground prong
- Ability to open plug for replacement
- Replaceable fuse to prevent faults
    - Potentially will use a capacitor instead for a low pass filter
- Wiring to ensure live wires are unplugged before ground once the cord is pulled out
    - On the inner part of the plug, the live wires will be straighter than the ground so that the live wires lose connection first
- Polarized plugs

Chase Nguyen

1/24/2025

Stage of Engineering Design Process: Prototype

When I got home, I began 3D printing our prototype for the plug. I used my Bambu X1C 3D Printer with PLA We used PLA filament because it is the basic material choice for 3D printed plastic, and this is just a prototype. For our actual product, we intend to use the general materials normal plugs typically use. Our TPU has not arrived yet, so I had to print the entire thing out of PLA, so it is closer to a model than a prototype.

We had to enlarge the size of our print because of the size of the nozzle, the nozzle is too big to print the incredibly thin and small covers. The wires would be able to fit through our design without being too cramped with the extra features we added. We will have to alter and tinker with the tolerances to ensure the parts fit well and are easy to replace.

Coby Nguyen

1/29/2025

Stage of Engineering Design Process: Prototype

We talked to our engineering teacher Mr. Hernandez to see how we would go about printing our parts without having to upscale the prototype model. He recommended that we use a resin 3D printer, which is much more precise and could print the small size of our plug.

For now, we have decided to increase the size of the covers, so that they are now .5mm thick. The printer is now able to print all the parts to scale without errors.

In the future, we want to use the TPU filament we ordered so that we can have a good amount of durability and flexibility in our cover and plug with the rest of it being ABS material for high durability and UV resistance (besides the metal prongs and wiring of course)

Nathan Viejo

1/30/2025

Stage of Engineering Design Process: Communication and Outreach

For our industrial letter and review, along with getting feedback and advice for the direction of our project, we decided to contact the CEO of LA Construction, a construction company.

Dear Team,
I am writing to commend you on your innovative work on Project ELECTROJAN. Your dedication to improving electrical safety through the redesign of B plugs is both timely and impactful. The construction and industrial sectors face significant challenges with electrical safety, and your project addresses critical gaps in current standards.
Your thorough research and thoughtful design process demonstrate a deep understanding of the hazards associated with electrical systems, particularly in high-risk environments such as construction sites and industrial facilities. The integration of features like retractable insulating prong covers, LED grounding indicators, and polarization not only enhances safety but also improves usability, a combination that is often difficult to achieve.
I was particularly impressed by your attention to real-world applications. The inclusion of a fuse mechanism like the United Kingdom's Type G plug is a cost-effective and practical solution for preventing overloads and short circuits. Additionally, the use of durable materials and the exploration of modular designs for industrial-grade outlets show a clear understanding of the demanding conditions these products must withstand.
Your focus on mitigating risks such as accidental shocks and arc flashes closely with the priorities of our industry. Construction sites would benefit greatly from your proposed innovations, as they often deal with exposed wiring, moisture, and heavy equipment. The potential to reduce electrical injuries and fatalities through your designs is important, and I am confident that your work could set a new standard for electrical safety in the U.S.
I encourage you to continue refining your prototypes and exploring partnerships with manufacturers and industry stakeholders. I believe that the concerns you are addressing are even more critical in industrial environments than in residential settings. Your project has the potential to not only improve safety but also drive innovation in electrical design across multiple sectors. If you are open to collaboration, our firm would be interested in supporting further development and testing of your designs in real-world construction environments.
Once again, congratulations on your outstanding work. Your project is a testament to the power of engineering to solve real-world problems and save lives. I look forward to seeing how Project ELECTROJAN evolves and impacts the industry.

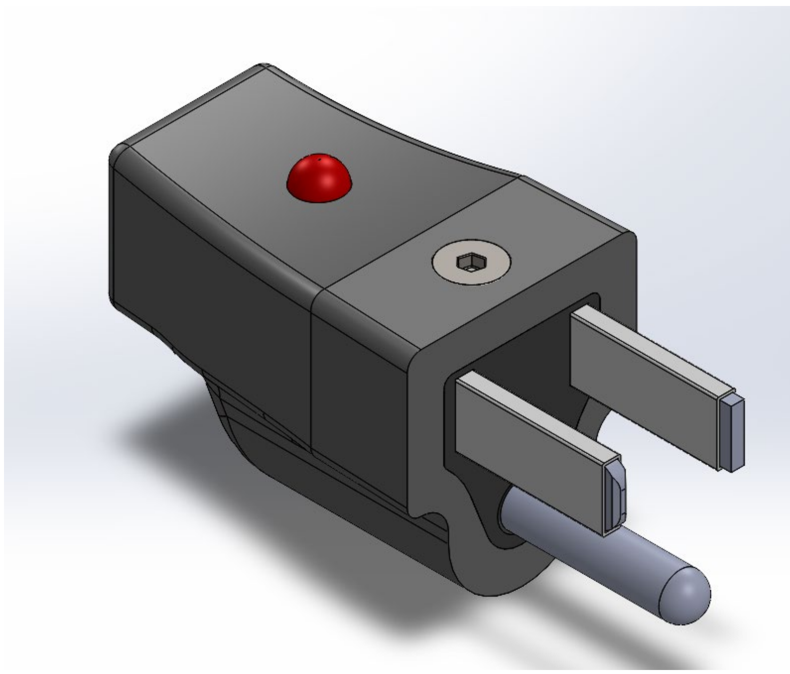
Best Regards,
Luis Arriaga
CEO
Grand Prairie, TX 75052
713-979-8106

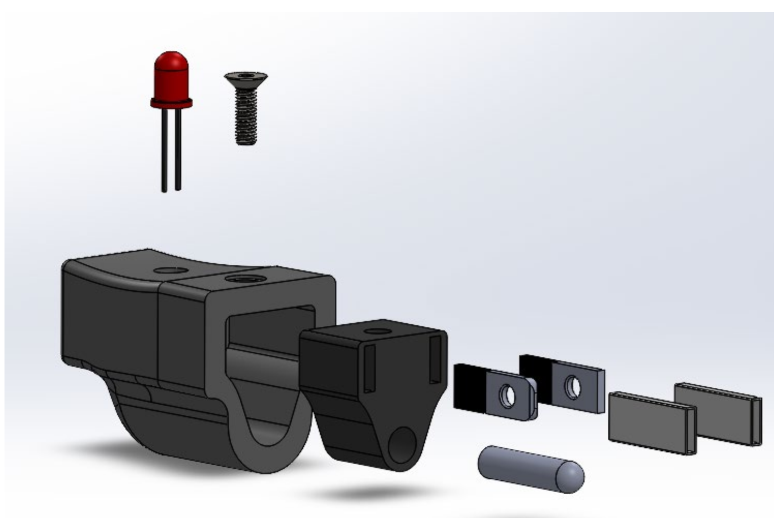
Figure 1. LA Construction's response to our project

2/02/2025

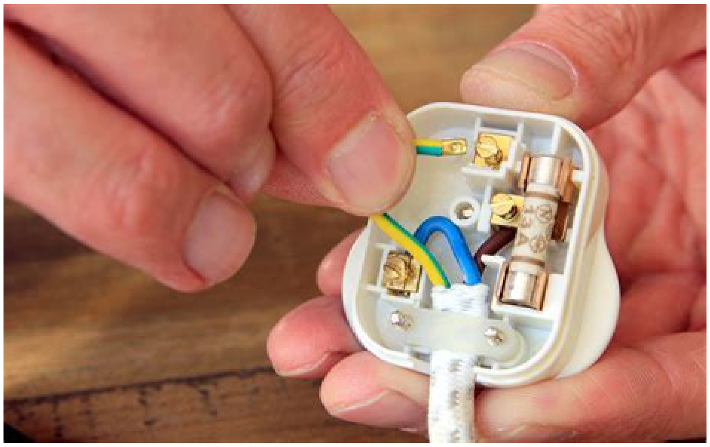
Stage of Engineering Design Process: Design

To ensure that our design is modular and simple to replace, we added some more features to our design. The main addition is a screw that screws into the outer and inner covers. This makes it so that you can take the inner cover from the front by unscrewing the top, so that you may access the rest of the plug.





The main design we plan to implement in the future is to make the design more similar to the UK Type G plug, or the replacement plugs you can find online. These plugs feature screwable covers for the plug, rather than the closed molds most Type A and B plugs feature, which make it difficult to replace and impossible to open without cutting them.



We may need to change the overall design of our plug, making it larger so that there is more space to optimize wiring like the Type G plug, and to ensure room for the fuse.

Coby Nguyen

2/02/2025

Stage of Engineering Design Process: Prototype

Since the TPU finally arrived, we were able to do a test print of our model. We sliced it for our teacher's printer, but the covers were not able to print due to their small size. The weird thing is that my slicer allows it to be sliced, but our teachers didn't let him. We decided to just print that part at home, this print mainly being a test anyways.

The TPU filament we bought is rated 95A, which is on the higher side of flexibility. The print turned out perfectly and was very squishy and satisfying to play with.

Figures 1 and 2: 3D print of plug design in TPU

Coby Nguyen

2/07/2025

Stage of Engineering Design Process: Research

To ensure proper understanding of plug design, we and our teacher Mr. Head cut open a nonfunctional plug to see how it is wired. The wiring can differ a bit depending in the manufacturer of course, but most of them are similar. The wiring is quite straightforward, all the wires connect directly to the prongs from the cord with insulation around it. There were no gaps or space, everything was filled with rubber insulation material.

Figure 1. Plug before cutting

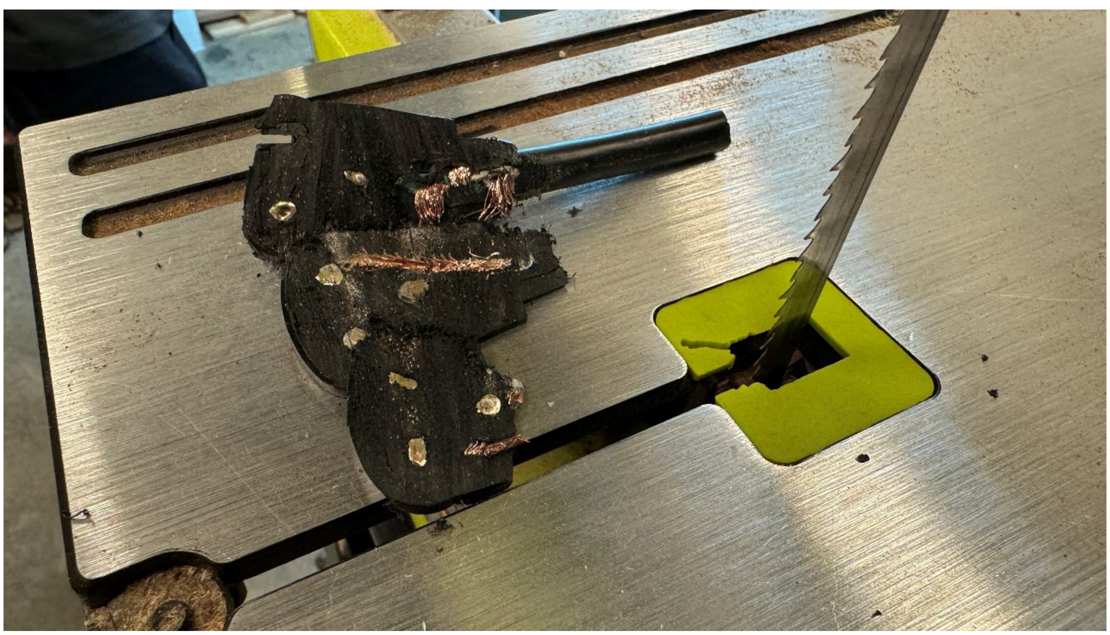


Figure 2. Each layer of the plug cut

Figure 3. One layer of the plug after being cut, showing the prongs and wires

Figure 4. Fully disassembled plug

Chase Nguyen

2/10/2025

Stage of Engineering Design Process: Research/Prototype

Today I used my time in my digital electronics class to research the wiring for our plug. A friend of mine in electrical engineering reviewed my initial circuit concept and provided some feedback. He emphasized that it would be beneficial to include detailed wiring and circuit diagrams to clearly show how the LED indicator would source its power without interfering with the primary device. His insight was that the LED could be powered by diverting a small current through a resistor connected in parallel, which would ensure that the LED lights up only when the plug is securely engaged, all while drawing minimal power.

He also said that our circuit needs to account for variable voltage and current conditions without becoming parasitic. I had my friend Trevor help me with exploring methods to make this work. There are a variety of methods we thought of, such as using an inductor to stabilize the current and keep the LED from flickering. This theoretically should work as inductors can resist the constant change in current.

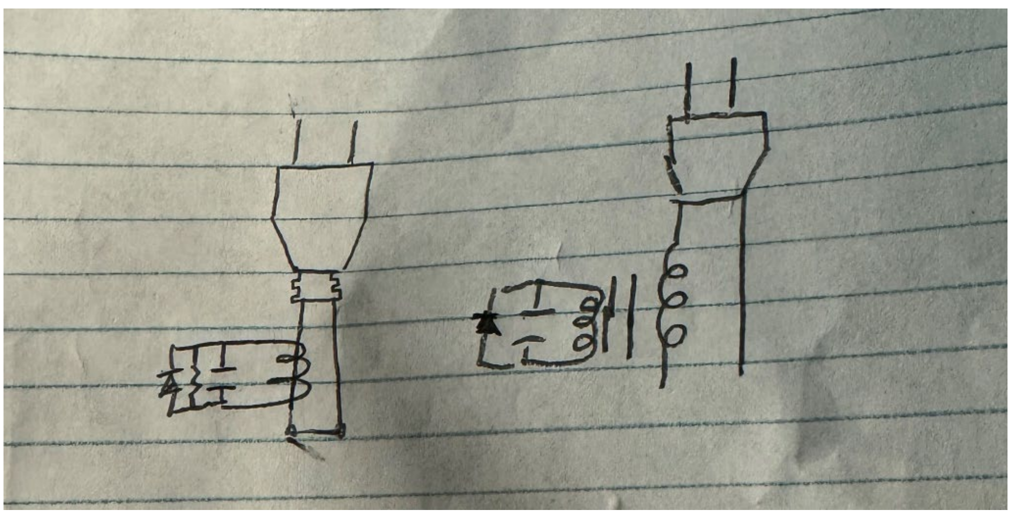


Figure 1. Sketch of plug using an inductor, capacitor, and resistor

Coby Nguyen

2/12/2025

Stage of Engineering Design Process: Design/Prototype

We conducted further research into the circuit design for our plug to ensure the LED can be powered safely without concerns about high current, high voltage, or parasitic draw. Our design utilizes a low-pass filter to stabilize the current supplied to the LED, filtering out high-frequency noise and preventing flickering.

The low-pass filter works by allowing low-frequency signals to pass through to the LED while blocking high-frequency signals. This is achieved using a combination of a resistor and a capacitor. The resistor limits the current, preventing high current from reaching the LED, while the capacitor stores and releases energy gradually, smoothing out any voltage spikes. As a result, only the steady, low-frequency components power the LED, ensuring consistent brightness without affecting the main device's power supply.

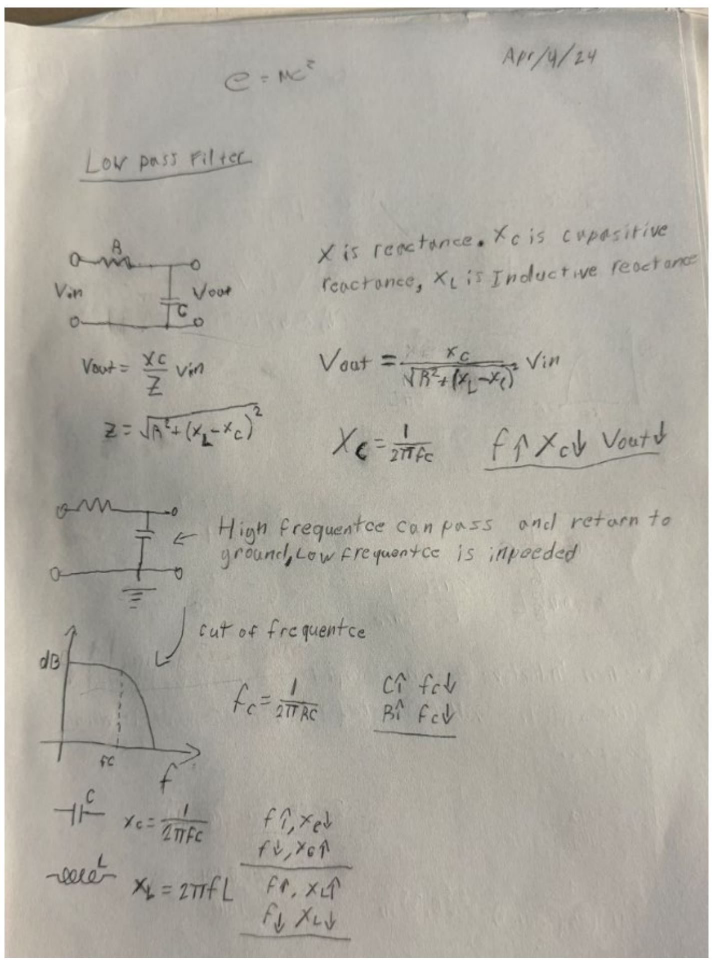


Figure 1. Our sketch of the filter

Chase Nguyen

2/18/2025

Stage of Engineering Design Process: Prototype/Testing

Today we worked on putting together our circuit using the components we have. First, we made our rectifier in Tinkercad so that we could simulate the circuit and figure out the calculations for our capacitor and resistor. Right now, we plan to use a 100k ohm resistor with a 10uF capacitor. This rectifier is paired with a low pass filter by limiting the circuit to lower frequencies whereas the rectifier converts AC to DC with the diodes.

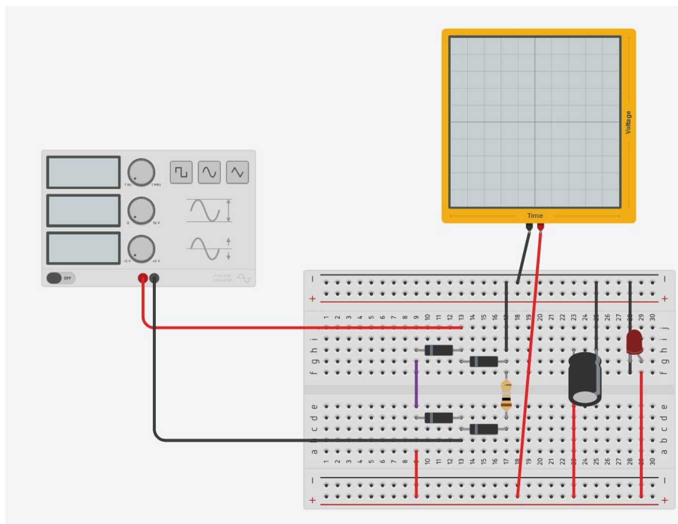


Figure 1. Full wave rectifier with low pass filter connected to LED

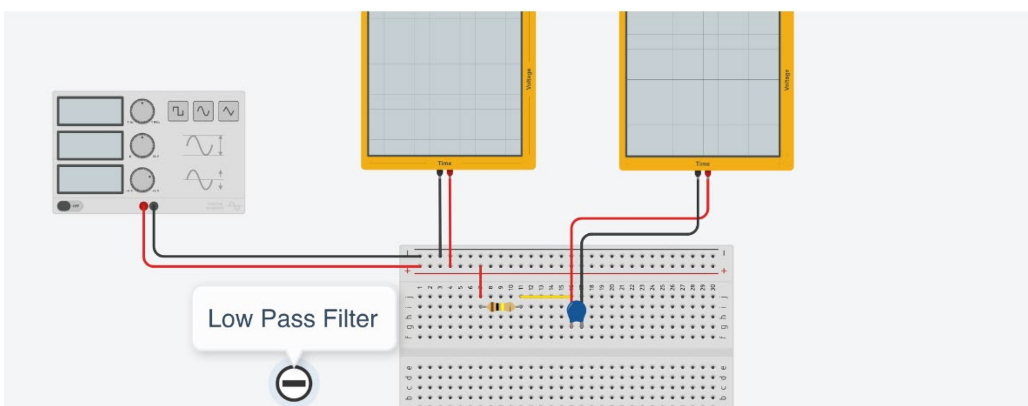


Figure 2. Low pass filter

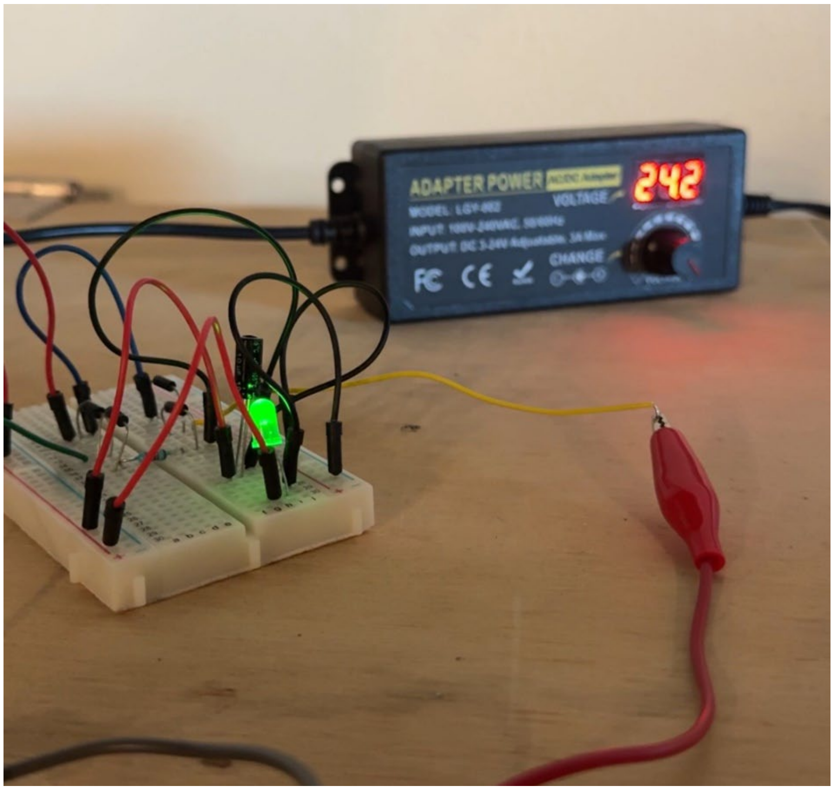


Figure 3. Working prototype of rectifier with breadboard at 24v DC

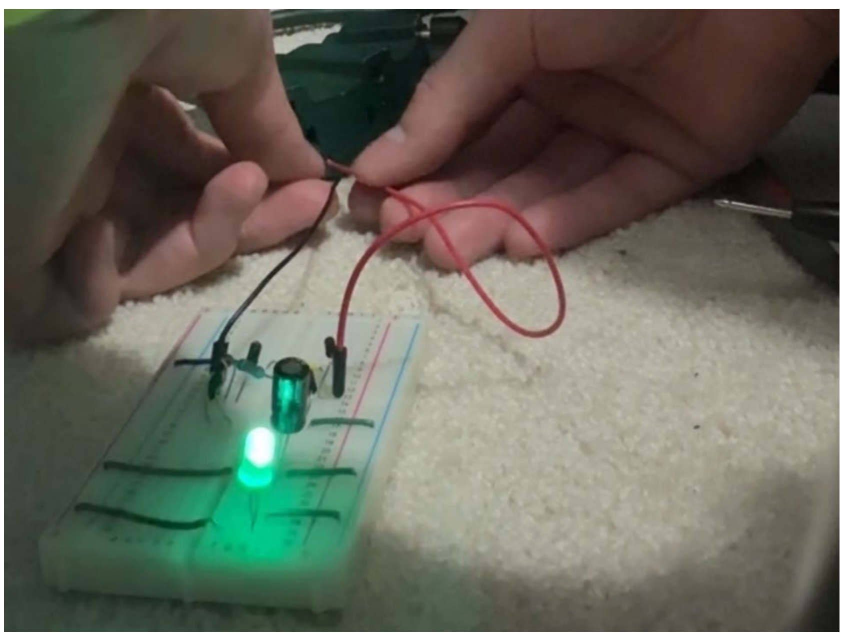


Figure 4. Working prototype of rectifier circuit with 120v AC

2/24/2025

Stage of Engineering Design Process: Brainstorm

Now that district competitions are finished, we have had time to brainstorm and upgrade our design. The main component we want to work on is upgrading the capabilities of our components so that they can take higher loads, and to work on the prong covers for our plug. We want to ensure that the prong covers have a nice fit over the prongs without being too tight. To do this we need an alternative method to connect the prongs to our outlet since surrounding them would be the covers, so we need to connect them to the prongs through the back.

We decided that we will likely need to increase the size of our plug so that more components can fit without it being crowded.

Chase Nguyen

3/12/2025

Stage of Engineering Design Process: Design

To get more accurate and efficient numbers in our circuit, we decided to switch from Tinkercad circuits to LTSpice, which is popular and open-source circuit simulation software. This will allow us to properly simulate alternating current without glitches, and without having to risk our components or safety by testing in real life before we are sure our circuit will properly function. The circuit will run parallel to the main device connected to the plug, which prevents any major voltage drop.

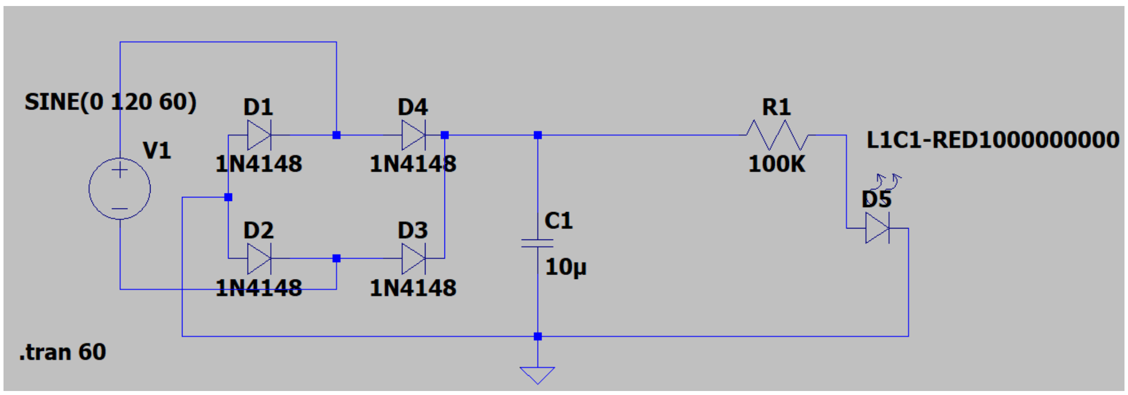
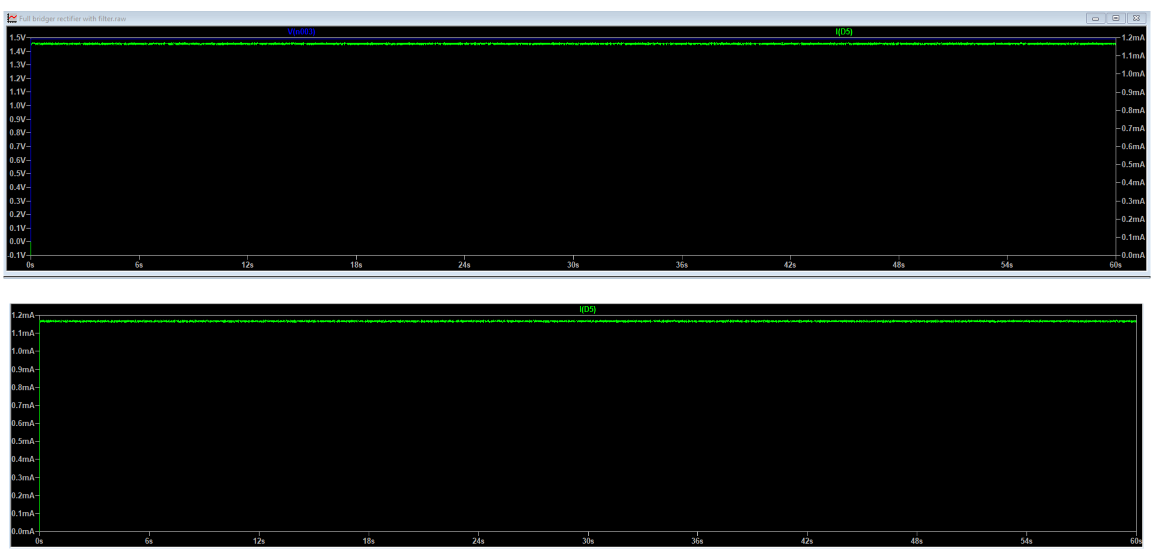


Figure 1. Circuit diagram of the LED indicator

Using the simulation features of LTSpice, we were able to check the voltage and current levels throughout the circuit. The LED receives 1.5V with 1.2mA. This meets the ratings of the LED by the manufacturers.



Figures 2 and 3: Graph of the voltage and current through the LED

Coby Nguyen

3/14/2025

Stage of Engineering Design Process: Research

Today, we collaborated and collected research on US's electrical system and how powered appliances are not used at 15 amps or 120 volts. As it is thought that everything runs on 120 volts evident on multimeter readings, circuits use 240 volts which makes outlets read 120 volts from power transformers twice, making 240. In summary, we can find variation with 120-240 volts that are around us, as 120 volts is primarily for regular household appliances like lighting and standard outlets, and 240 volts for high-powered appliances like ovens, dryers, and air conditioning.

Technology Connections. *The US electrical system is not 120V*. YouTube, 22 Jun. 2020, https://www.youtube.com/watch?v=jMmUoZh3Hq4

We also decided to do research on faults identified within our electrical system here in the US that mainly puts the focus on extension cords. Problems can also be found in breakers as well when a circuit is overloaded or the breaker itself is worn, and even the breaker itself can fail to trip when needed, or trip unnecessarily. The way we plan to counter these issues found in different electrical components is through our further continuation of our design process to fully ensure our plug is protective for the human body.

Technology Connections. *Perhaps the weakest link in the US electrical system*. YouTube, 30 Jul. 2021, https://www.youtube.com/watch?v=K_q-xnYRugQ

Chase Nguyen

3/17/2025

Stage of Engineering Design Process: Research

For today, we did further research on the design aspect of US-styled outlets. Within our research based on the video, we saw flaws within how the outlet is oriented within a wall which what has been done for change is inverting the outlet to where the ground prong plug is at the top, and the plug for the two prongs are at the top, which has been said to help reduce risks of electrical shock as the ground wiring is on the top as a way of protection where there's no electricity passing through. The video also included on how "GROUND" and "TEST" are labeled for GFCI outlets which reads facing either way as the face of the outlet can be installed either way. Another key point mentioned in the video was over plugs being partially inserted in outlets which can be dangerous as it leads into electrical shock as there are no insulation sleeves from electrical faults, demonstrating the key focus of our project with our prong-covering mechanism

Technology Connections. *Power outlets are topsy turvy - but does it matter?*. YouTube, 26 Sep. 2023, https://m.youtube.com/watch?v=vNj75gJVxcE

We also collaborated for another video that we decided to do research over that covers GFCI outlets. These plugs are helpful as they protect against the occurrence of electrical shocks and potentially sparked fires, as these outlets can rapidly detect and interrupt electrical current leaks to ground in problems such as faulty wiring and malfunctions in the source of connection. On the GFCI outlets, two buttons can be found which "TEST" is for a daily check to see that the GFCI outlet properly works, and "RESET" being for once the test has been done and restored power or current to go through the outlet.

Technology Connections. *The GFCI/RCD: A Simple but Life-Saving Protector*. YouTube, 17 Aug. 2018, https://www.youtube.com/watch?v=ILBjnZq0n8s

Nathan Viejo

3/18/2025

Stage of Engineering Design Process: Design

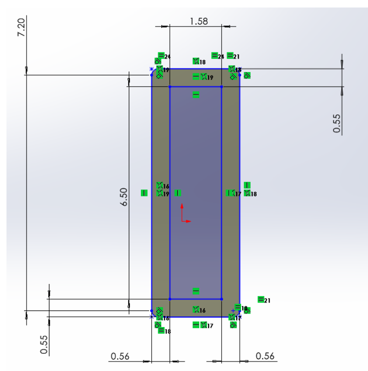
To ensure that the retractable covers properly slid in place, we altered the tolerance of the holes for our 3D printer. Previously, even with .2mm tolerances, it was too tight of a fit to slide into the prong covers. We tuned the tolerances to fit our Bambu X1C 3D printer and the .4mm nozzle.

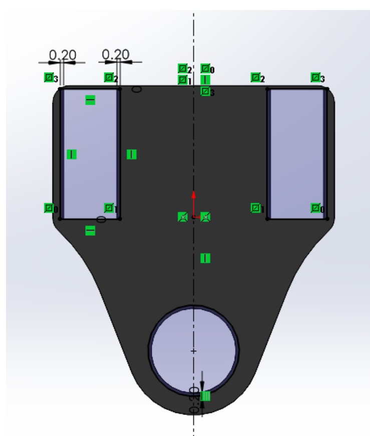
<u>Tolerances</u>

Outer cover to Inner cover: .2mm x .2mm

Cover to inner cover: .26mm x .13mm

Cover to prongs: .14mm x .24mm





Figures 1 and 2: Tolerances set for the prong cover and inner cover

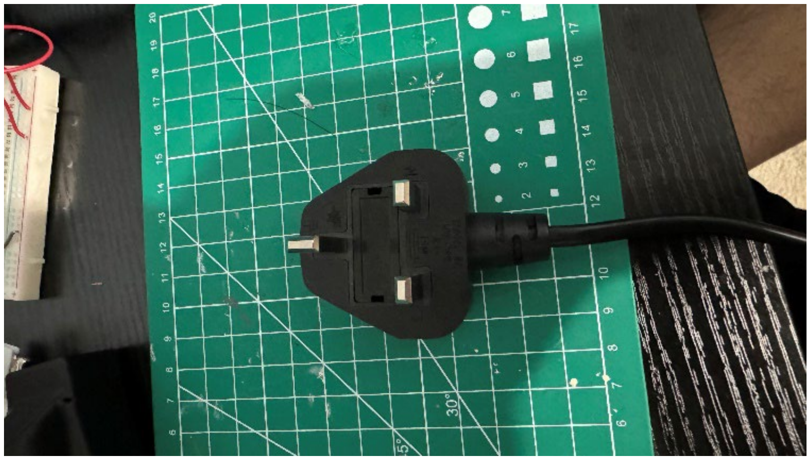
Coby Nguyen

3/22/2025

Stage of Engineering Design Process: Prototype/Testing

We printed this prototype with black PLA for testing purposes. It took around 30 minutes to print, 1 hour total for both plates. The tolerances work perfectly, with a not too much friction which allows for smooth sliding.

Figures 1-2: 3D Printed model of enlarged plug design



Figures 3-5: UK Type G plug with fuse

3/24/2025

Stage of Engineering Design Process: Design

Today we worked on improving the design so that the plug allows for the prongs to be held in place. Previously, they were floating since they could not be attached from the sides since the covers were surrounding them, and the covers could not be connected since they must slide back and forth.

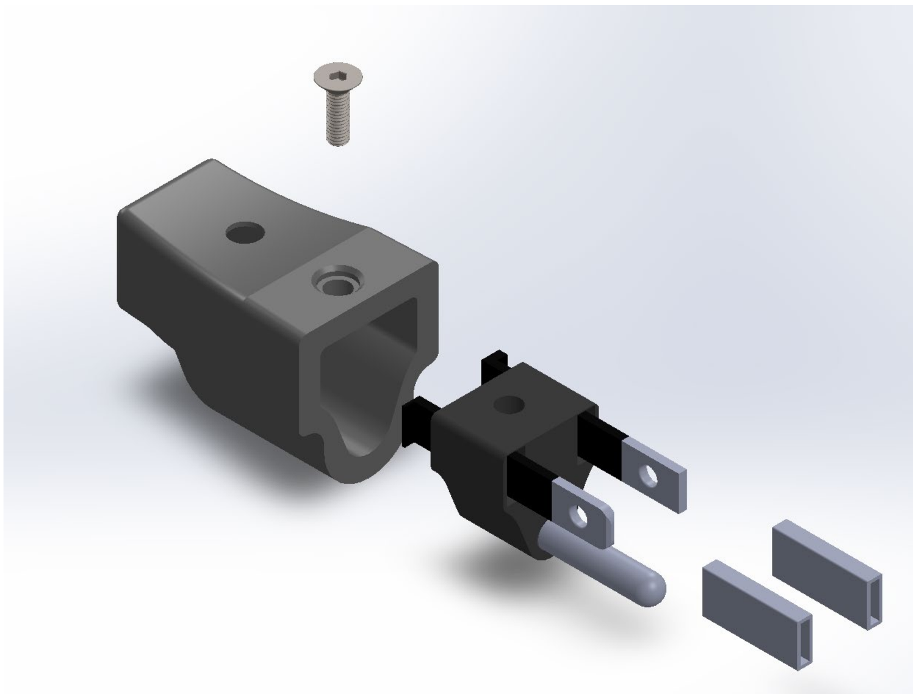


Figure 1: Exploded view of assembly

Nathan Viejo

3/26/2025

Stage of Engineering Design Process: Design

Today we started coming up with how the spring mechanism will be made, which we plan on making the prongs longer throughout the inside of the plug. This will allow for a base within the plug for the spring to control the compression and tension of the spring as the plug is being inserted and taken out of an outlet.

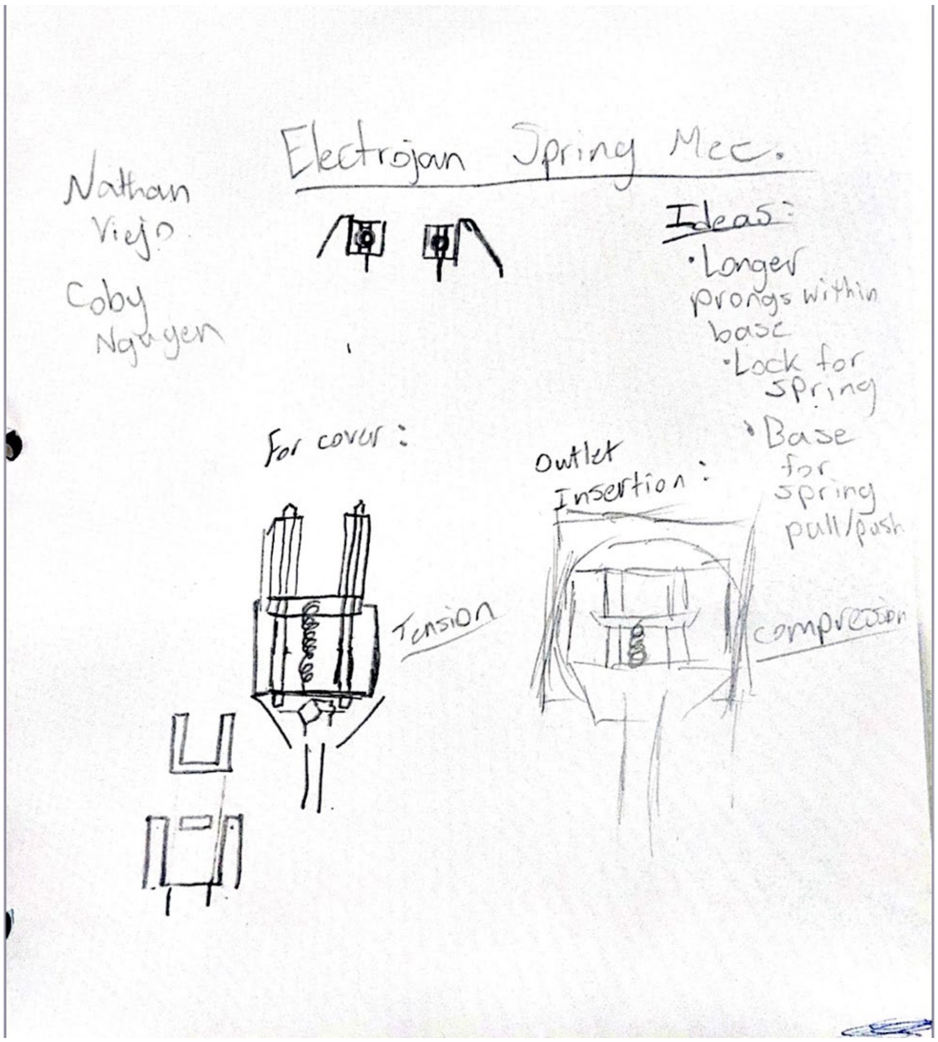


Figure 1: Sketch of spring mechanism

Chase Nguyen

3/27/2025

Stage of Engineering Design Process: Testing

Today, we conducted experiments using an oscilloscope and a function generator to better understand their operation and functionality. These devices will play a key role in our demonstration, serving as the power supply for our plug.

The function generator produces an AC signal at a specific frequency and amplitude, simulating the electrical flow that would normally come from a power source. By adjusting its settings, we can control the voltage and waveform, allowing us to replicate real-world electrical conditions.

The oscilloscope is used to visualize this electrical signal, displaying the waveform on a screen in real-time. This allows us to observe key characteristics such as voltage levels, frequency, and signal stability. In our demonstration, the oscilloscope will show the alternating current waveform when Electrojan is fully connected, confirming that power is flowing through the system.

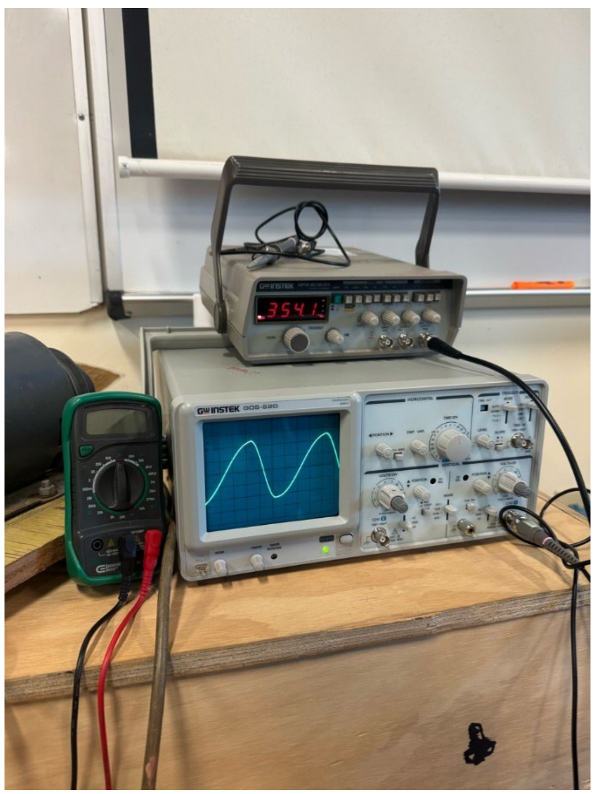


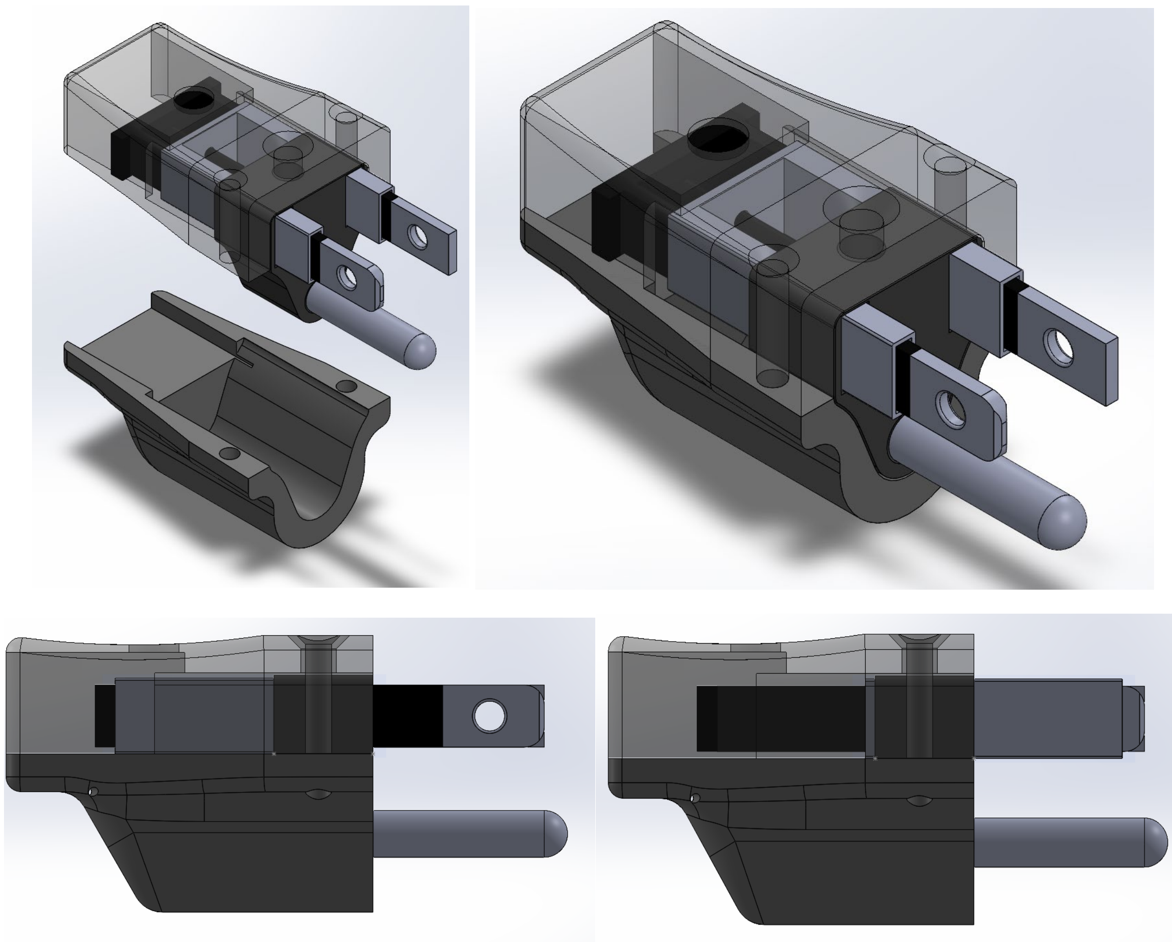
Figure 1: Function generator (top) and oscilloscope (bottom)

Chase Nguyen

3/30/2025

Stage of Engineering Design Process: Design

Today we worked on the design, improving modularity. We decided to divide the outer shell into a top and bottom half, allowing people to easily assemble it and replace the parts. This also solves the problem we had with keeping all the parts in place, as they will not be able to move (besides the prong covers).



Figures 1-4: View of current design of plug with prong covers retracted and extended

Coby Nguyen

3/31/2025

Stage of Engineering Design Process: Prototype

Today we focused on printing the prototype so that we could test tolerances and sizes. We had to make several adjustments to the printer settings and tolerances until we got the perfect fit for the print.

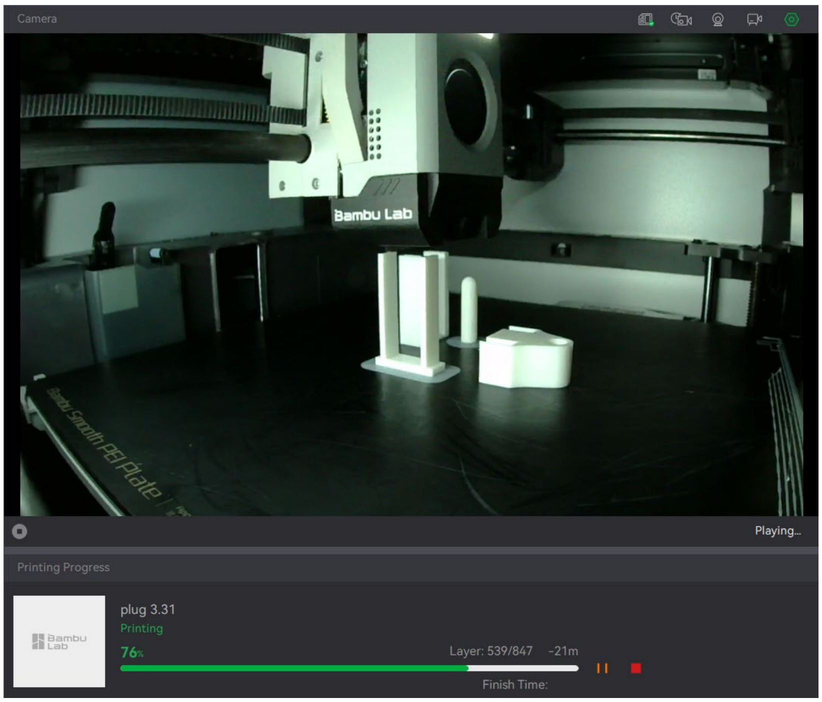


Figure 1: Bambu X1C printing the prongs, covers, and inner cover

Coby Nguyen

4/01/2025

Stage of Engineering Design Process: Prototype/Testing

After testing the sliding mechanism to ensure that the tolerances are correct, we printed the rest of the assembly. We printed the top and bottom shells along with the inner covers. We used our teacher's printer so that we could quickly print and alter our design in class. We realized that we would need to shorten the length of our prongs and prong covers because the prongs are extruded very long beyond the actual plug which is not normally how it is supposed to look. Besides that, the new print of our plug is working flawlessly and is ready for competition.

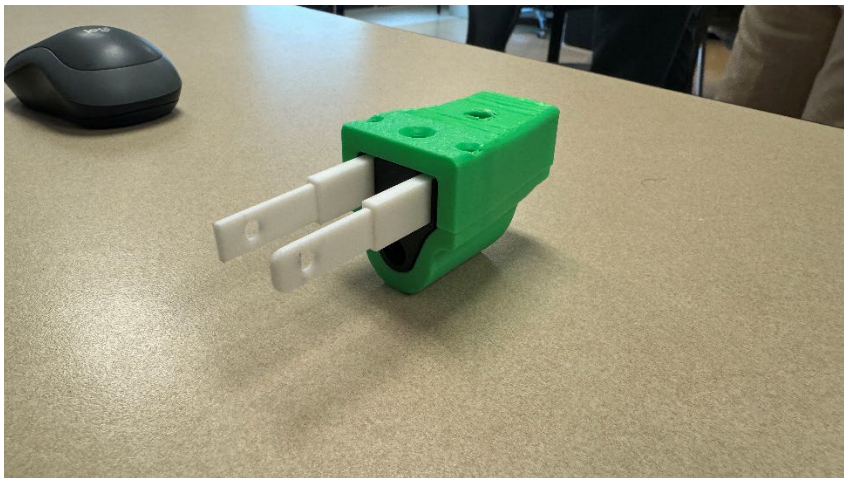


Figure 1: 3D Print of new shell design

Chase Nguyen

Industrial Letters

Dear Team,

I am writing to commend you on your innovative work on Project ELECTROJAN. Your dedication to improving electrical safety through the redesign of B plugs is both timely and impactful. The construction and industrial sectors face significant challenges with electrical safety, and your project addresses critical gaps in current standards.

Your thorough research and thoughtful design process demonstrate a deep understanding of the hazards associated with electrical systems, particularly in high-risk environments such as construction sites and industrial facilities. The integration of features like retractable insulating prong covers, LED grounding indicators, and polarization not only enhances safety but also improves usability, a combination that is often difficult to achieve.

I was particularly impressed by your attention to real-world applications. The inclusion of a fuse mechanism like the United Kingdom's Type G plug is a cost-effective and practical solution for preventing overloads and short circuits. Additionally, the use of durable materials and the exploration of modular designs for industrial-grade outlets show a clear understanding of the demanding conditions these products must withstand.

Your focus on mitigating risks such as accidental shocks and arc flashes closely with the priorities of our industry. Construction sites would benefit greatly from your proposed innovations, as they often deal with exposed wiring, moisture, and heavy equipment. The potential to reduce electrical injuries and fatalities through your designs is important, and I am confident that your work could set a new standard for electrical safety in the U.S.

I encourage you to continue refining your prototypes and exploring partnerships with manufacturers and industry stakeholders. I believe that the concerns you are addressing are even more critical in industrial environments than in residential settings. Your project has the potential to not only improve safety but also drive innovation in electrical design across multiple sectors. If you are open to collaboration, our firm would be interested in supporting further development and testing of your designs in real-world construction environments.

Once again, congratulations on your outstanding work. Your project is a testament to the power of engineering to solve real-world problems and save lives. I look forward to seeing how Project ELECTROJAN evolves and impacts the industry.

Best Regards,
Luis Arriaga
CEO
Grand Prairie, TX 75052
713-979-8106

LA
CONSTRUCTION LLC
Drywall and metal framing designs
Grand Prairie Tx 7139798106

Hello Team,

Thank you for involving me in the review of your innovative Electrojan plug design project. With my background in electrical safety and product development, I understand the critical role that plug, and outlet designs play in ensuring safe and efficient power connections. I really appreciate that the primary focus of your design is improving electrical safety, an essential concern in both consumer and industrial applications. Your design integrates several key safety features that significantly enhance standard plug designs. The inclusion of retractable prong covers and insulative material on prongs reduces the risk of accidental shocks and unintended contact with live connections.

Additionally, the LED indicator for secure connections provides a visual confirmation that a device is properly plugged in, helping to prevent loose or faulty connections that could lead to electrical hazards. The ability to open and replace components ensures long-term usability and makes maintenance more accessible. I also appreciate the research and attention to international safety standards in your design process. By analyzing plug types from different regions, such as Britain’s Type G plug with its built-in fuse for fault protection, you’ve incorporated elements that improve safety beyond what is commonly seen in U.S. Type A and B plugs. The decision to extend the ground prong and include holes in prongs for secure connections are small but impactful improvements that contribute to overall stability and safety.

Moving forward, I recommend ensuring that Electrojan complies with relevant electrical safety standards and regulations, such as UL 498 (Underwriters Laboratories for electrical plugs), NEC (National Electrical Code), and IEC 60884 (International Electrotechnical Commission standards for plugs and socket-outlets). Achieving these certifications will help validate the safety and reliability of your design for widespread use.

Overall, Electrojan is an exciting innovation that prioritizes safety, usability, and longevity in electrical plug design. I look forward to seeing how your project evolves and how it can contribute to safer electrical connections in homes and industries. Please feel free to reach out if you need any further feedback or assistance.

Best regards,
Christian Flores
ONCOR | Transmission Grid Operations

ONCOR

Dear Electrojan Team,

Thank you for sharing your Electrojan plug design with us. As a firm specializing in electrical design and construction, CEC Construction understands the importance of safety, efficiency, and adaptability in electrical components. We are always looking for innovations that enhance electrical safety and usability, and we appreciate the thoughtfulness behind your design.

Your approach to integrating safety features directly into the plug itself is commendable. The retractable prong covers and insulative material on prongs effectively reduce exposure to live electrical components, addressing common hazards in both residential and commercial settings. The LED indicator for secure connections adds an extra layer of assurance, allowing users to quickly identify whether their connection is stable. The ability to open and replace components ensures that maintenance and repairs can be done efficiently without needing a full replacement.

We also recognize the depth of research that has gone into this project. Studying international plug designs, particularly Britain's Type G plug with its built-in fuse, shows a commitment to enhancing safety beyond current U.S. standards. The decision to include a longer ground prong and holes for secure connections further supports stability and durability, especially in high-use environments.

From a regulatory standpoint, we strongly recommend ensuring that Electrojan complies with key electrical standards, including:

- UL 498 – Safety standards for electrical connectors and receptacles
- NEC (National Electrical Code) – U.S. guidelines for electrical installations
- IEC 60884 – International safety standards for plug and socket systems
- NEMA (National Electrical Manufacturers Association) ratings – Guidelines for durability and environmental resistance

We are excited about the potential of Electrojan and its ability to improve safety, reliability, and efficiency in electrical plug design. If there is anything our team at CEC Construction can do to assist in further development, testing, or compliance evaluation, please don't hesitate to reach out.

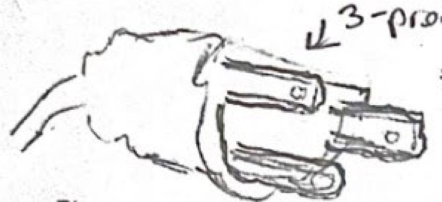
Best regards,
NAME
CEC Construction | Electrical Design & Engineering

CEC

Sketches

Electrojan - Planning

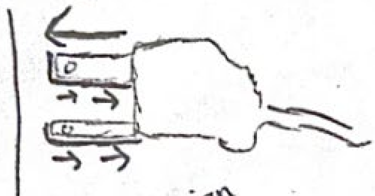
3-prong plug



Darker lining indicates Electrojan product

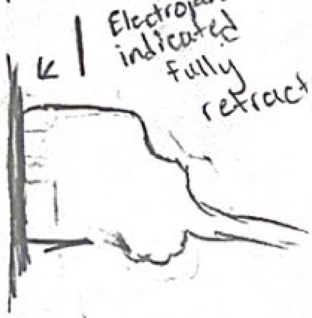
Material: Insulative coating
Non-conductive of electricity

Plug outlet



As prongs are being inserted into the outlet, Electrojan will retract backwards until prongs are completely inserted and electricity passes through

Electrojan indicated fully refracted

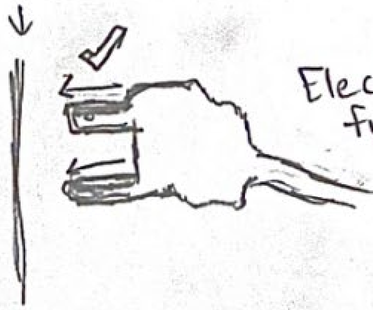


Plug outlet

Electrojan starts to cover prongs

In vise versa of outlet being removed, Electrojan will begin start to cover prongs as it leaves outlet to prevent the case of a shock if a finger makes contact with a prong

plug outlet

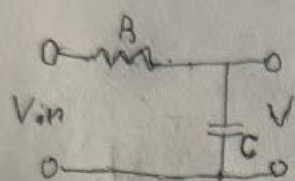


Electrojan is now fully covering the prongs as the plug is no longer inserted into the plug outlet

Apr/4/24

$e = mc^2$

Low pass Filter



X is reactance. X_C is capasitive reactance, X_L is Inductive reactance

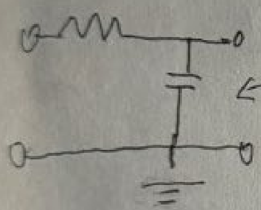
$$V_{out} = \frac{X_C}{Z} V_{in}$$

$$V_{out} = \frac{X_C}{\sqrt{R^2 + (X_L - X_C)^2}} V_{in}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

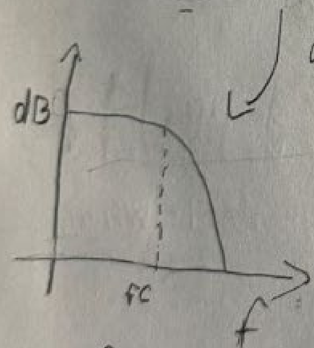
$$X_C = \frac{1}{2\pi f C}$$

$f\uparrow X_C\downarrow V_{out}\downarrow$



High frequentce can pass and return to ground, Low frequentce is inpoeded

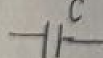
cut of frequentce



$$f_c = \frac{1}{2\pi RC}$$

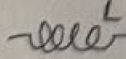
$C\uparrow f_c\downarrow$

$R\uparrow f_c\downarrow$



$X_C = \frac{1}{2\pi f C}$ $f\uparrow, X_C\downarrow$

$f\downarrow, X_C\uparrow$



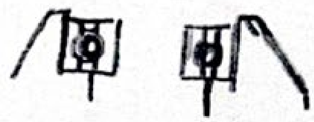
$X_L = 2\pi f L$ $f\uparrow, X_L\uparrow$

$f\downarrow X_L\downarrow$

Electrojan Spring Mec.

Nathan Viejo

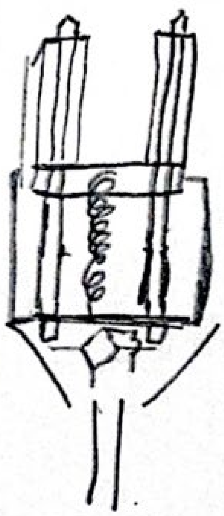
Coby Nguyen



Ideas:

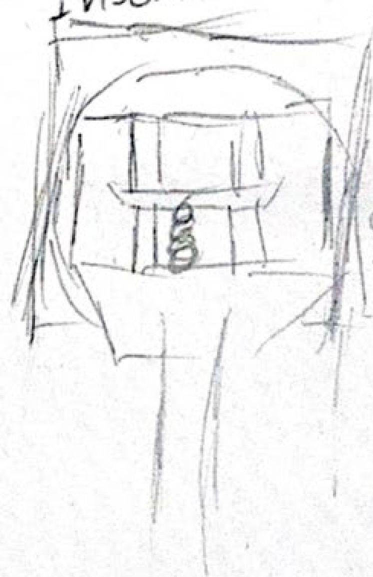
- Longer prongs within base
- Lock for spring
- Base for spring pull/push

For cover:

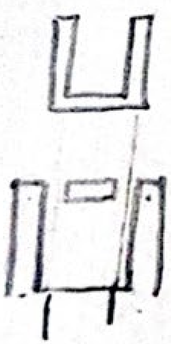


Tension

Outlet Insertion:

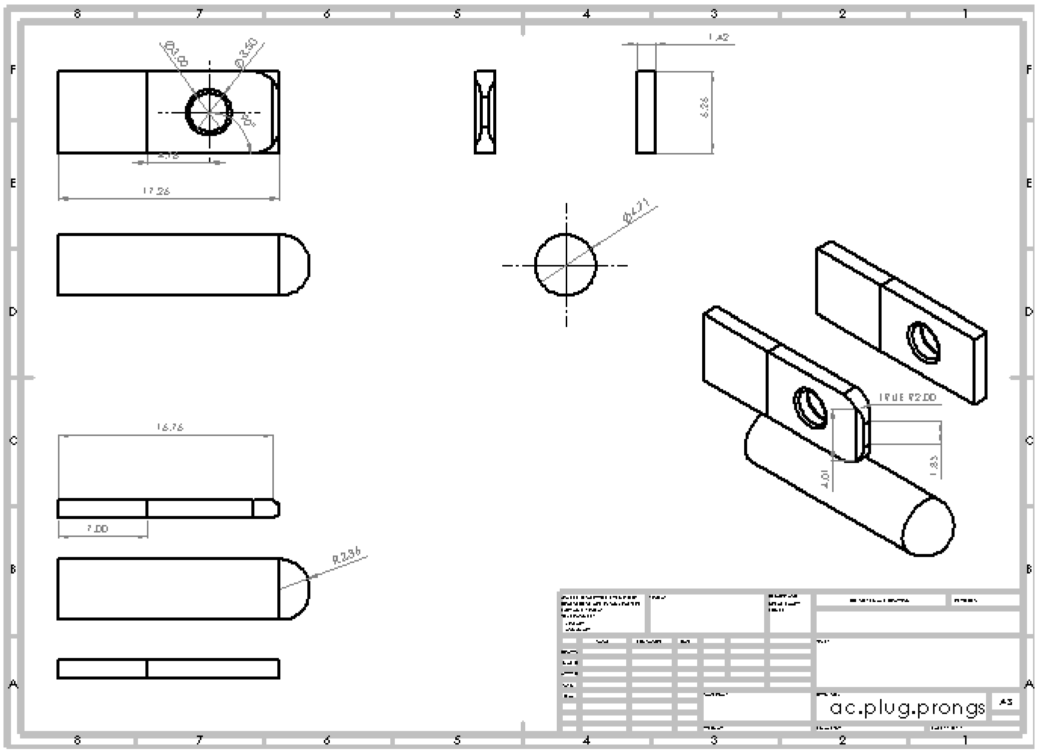


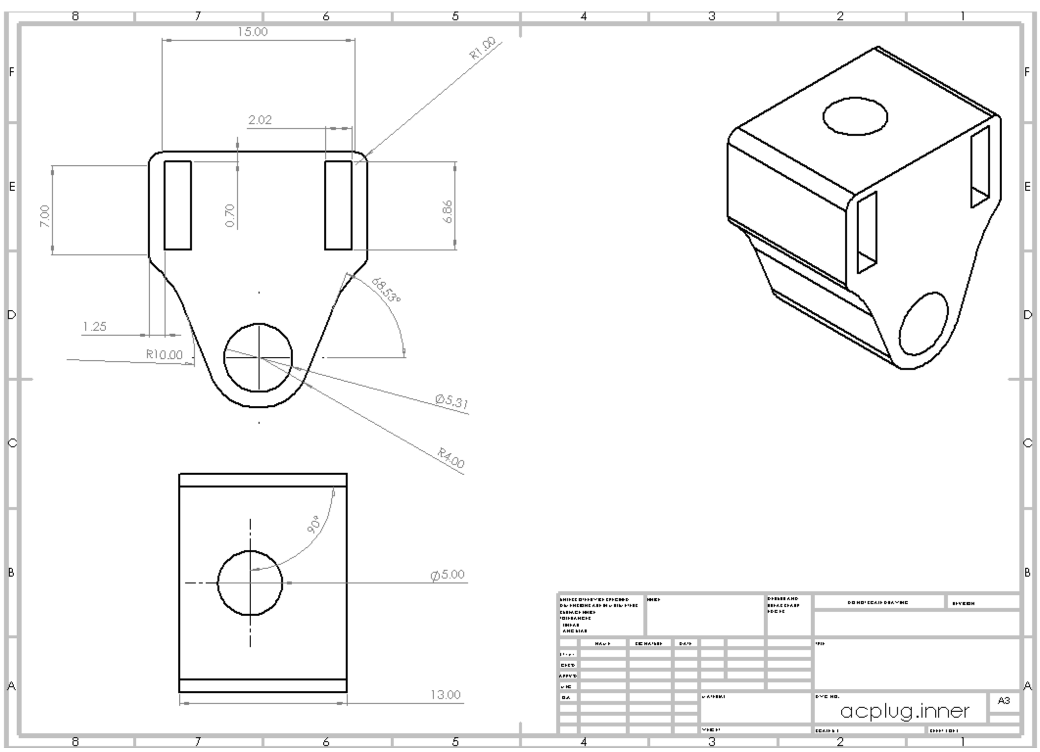
compression

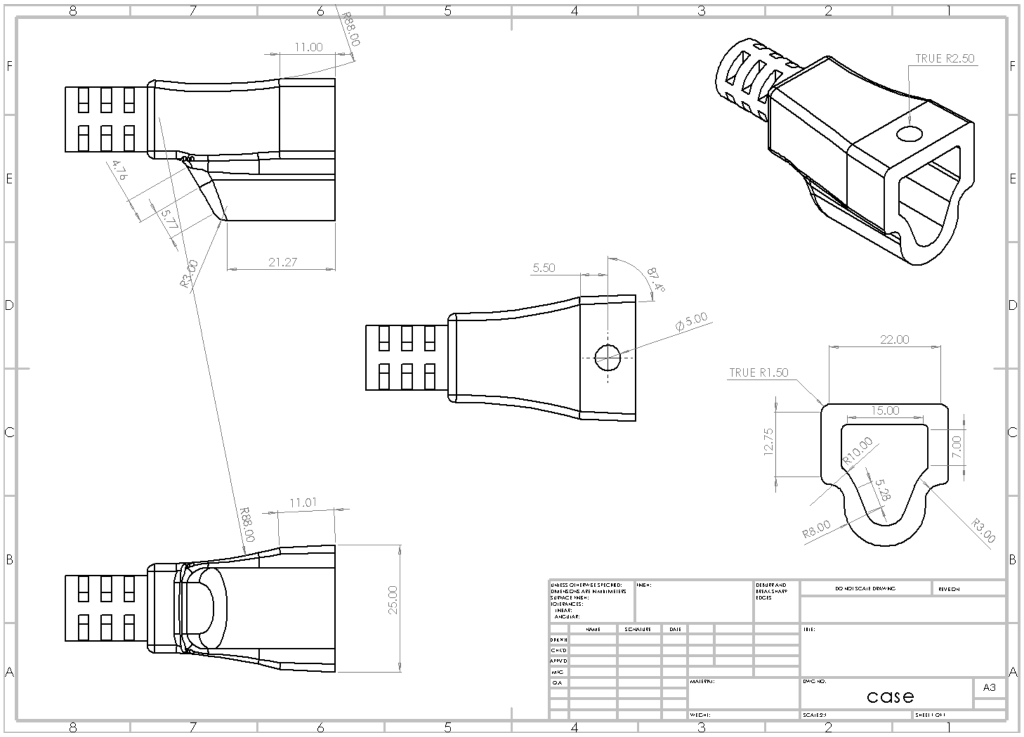


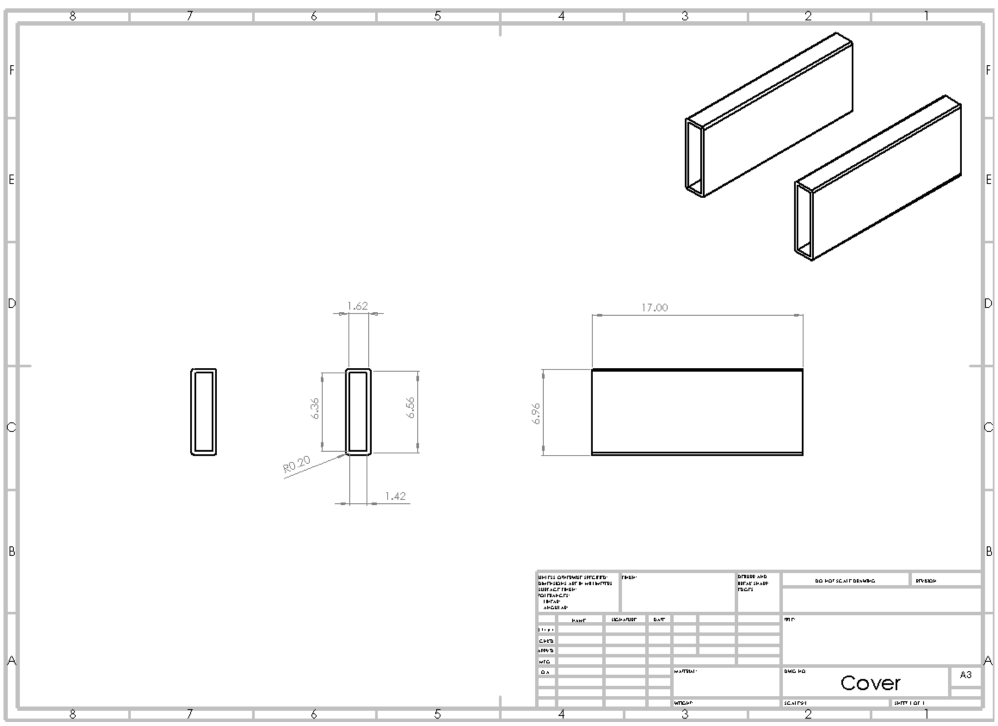
SolidWorks Designs

Version 1

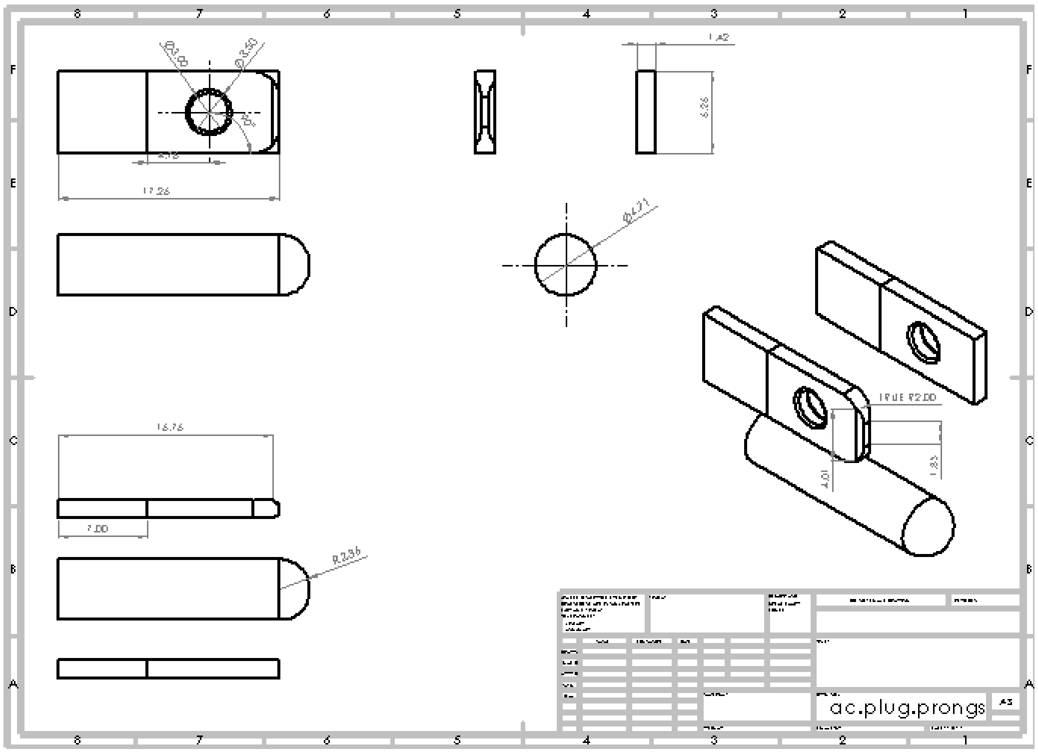

ac.plug.prongs

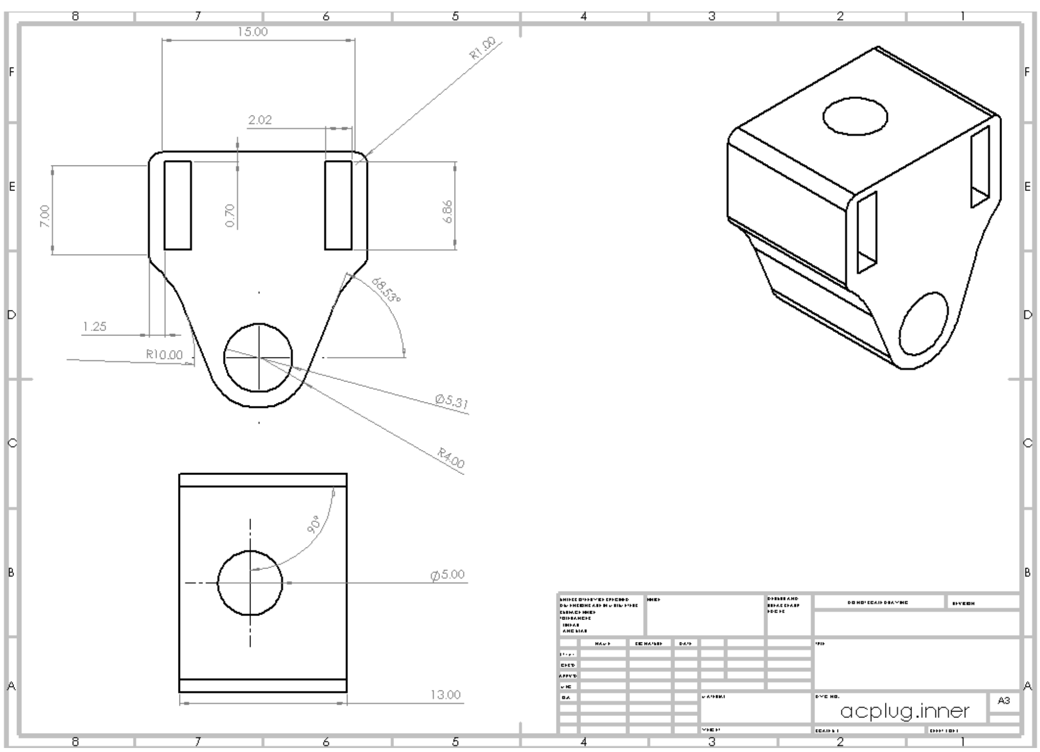

15.00
R1.00
2.02
7.00
0.70
6.86
68.53°
1.25
R10.00
Ø5.31
R4.00
90°
Ø5.00
13.00
acplug.inner
A3

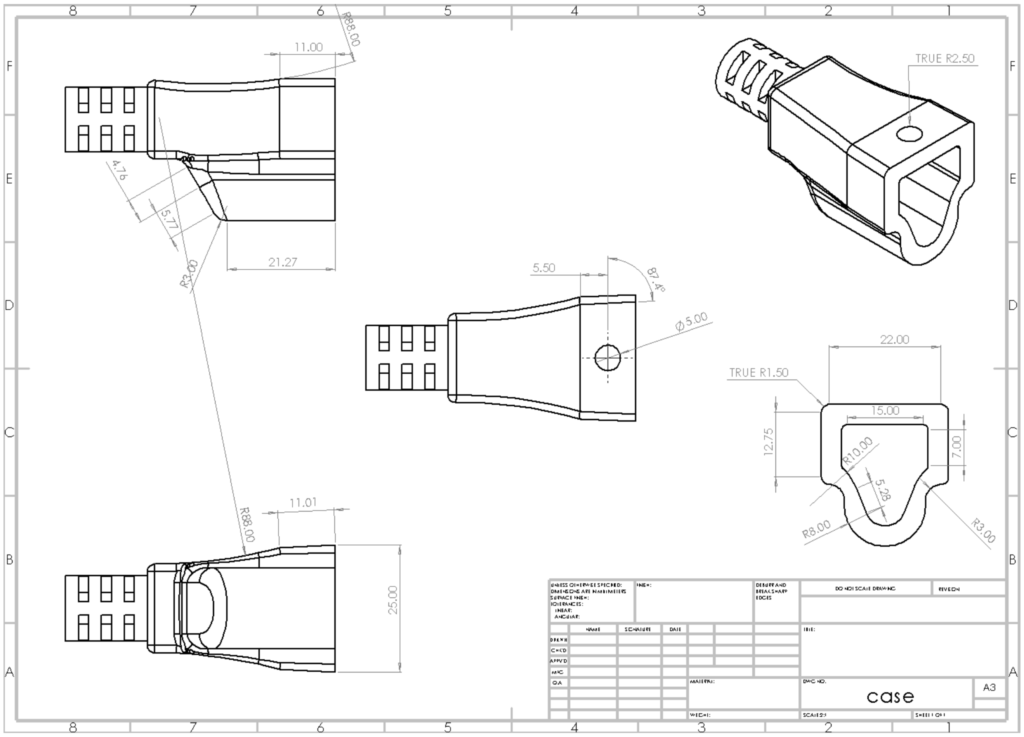

11.00
R88.00
TRUE R2.50
4.76
5.77
R3.00
21.27
5.50
87.4°
Ø5.00
22.00
TRUE R1.50
15.00
12.75
7.00
R10.00
5.28
R8.00
R3.00
11.01
R88.00
25.00
DO NOT SCALE DRAWING
case
A3

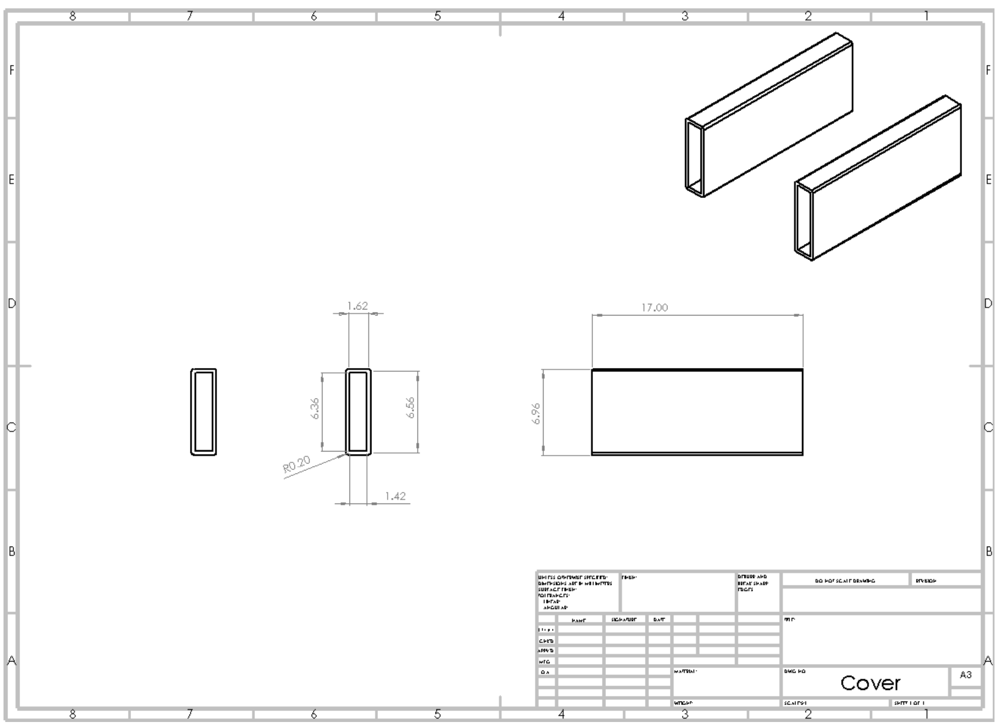

1.62
17.00
6.36
6.56
6.96
R0.20
1.42
DO NOT SCALE DRAWING
Cover
A3

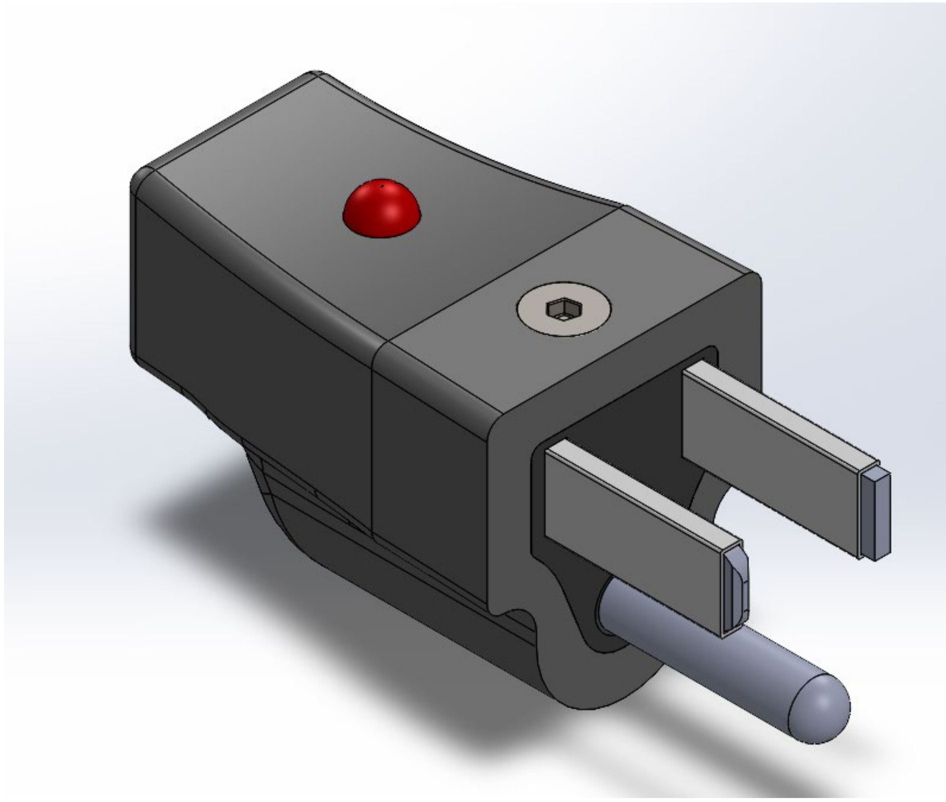
Version 2

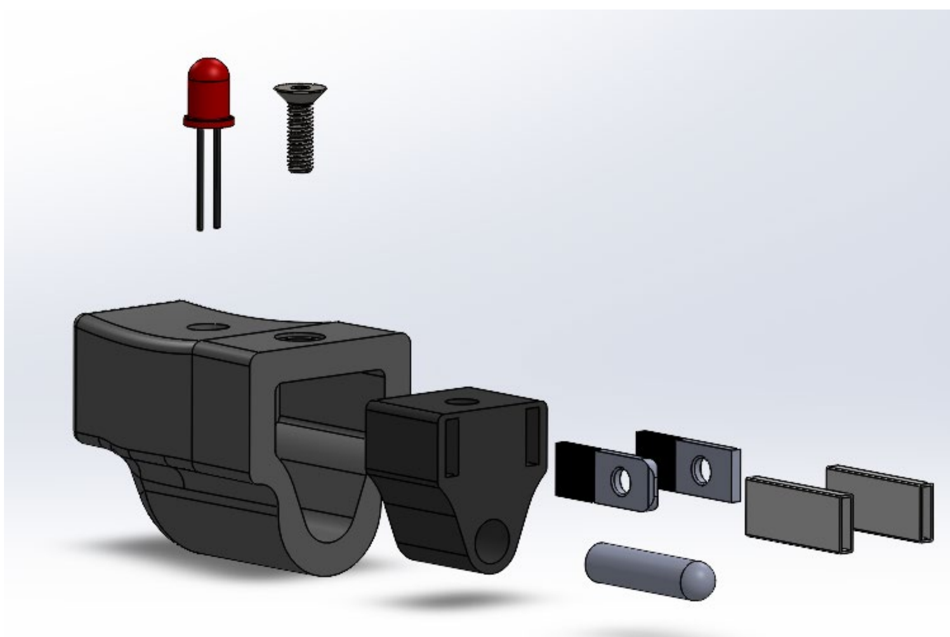

17.26
16.76
7.00
R2.36
6.26
TRUE R2.00
1.83
ac.plug.prongs
A3


15.00
R1.00
2.02
7.00
0.70
6.86
68.53°
1.25
R10.00
Ø5.31
R4.00
90°
Ø5.00
13.00
acplug.inner
A3

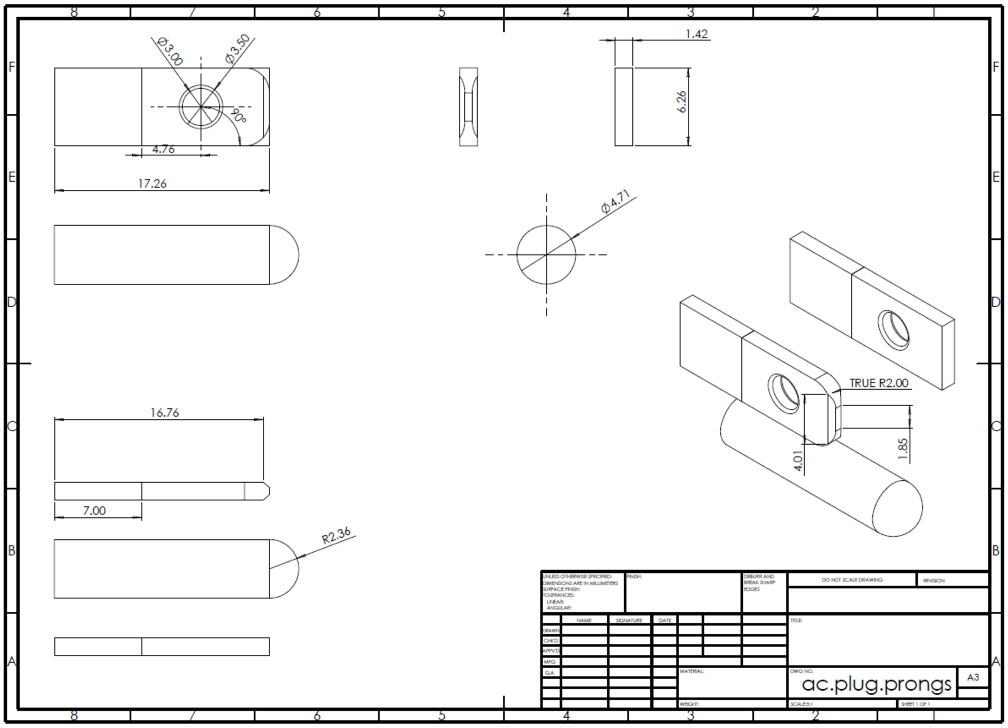

11.00
R88.00
TRUE R2.50
4.76
5.77
21.27
R3.00
5.50
87.4°
Ø5.00
22.00
TRUE R1.50
15.00
12.75
7.00
R10.00
5.28
R8.00
R3.00
11.01
R88.00
25.00
case
A3


1.62
17.00
6.36
6.56
6.96
R0.20
1.42
Cover
A3

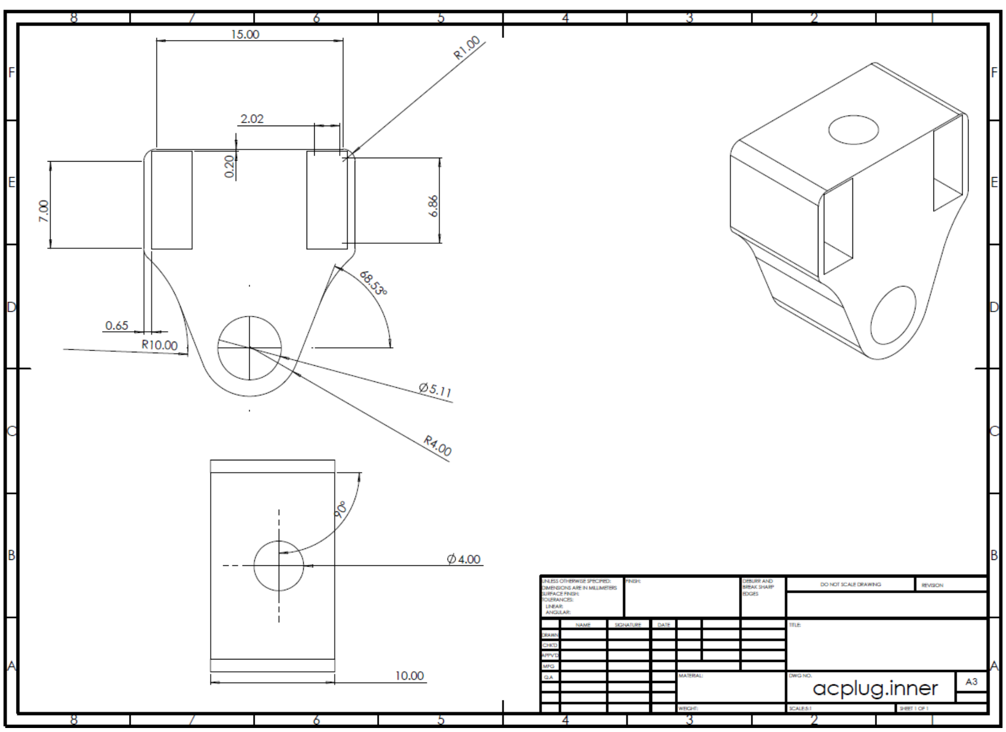




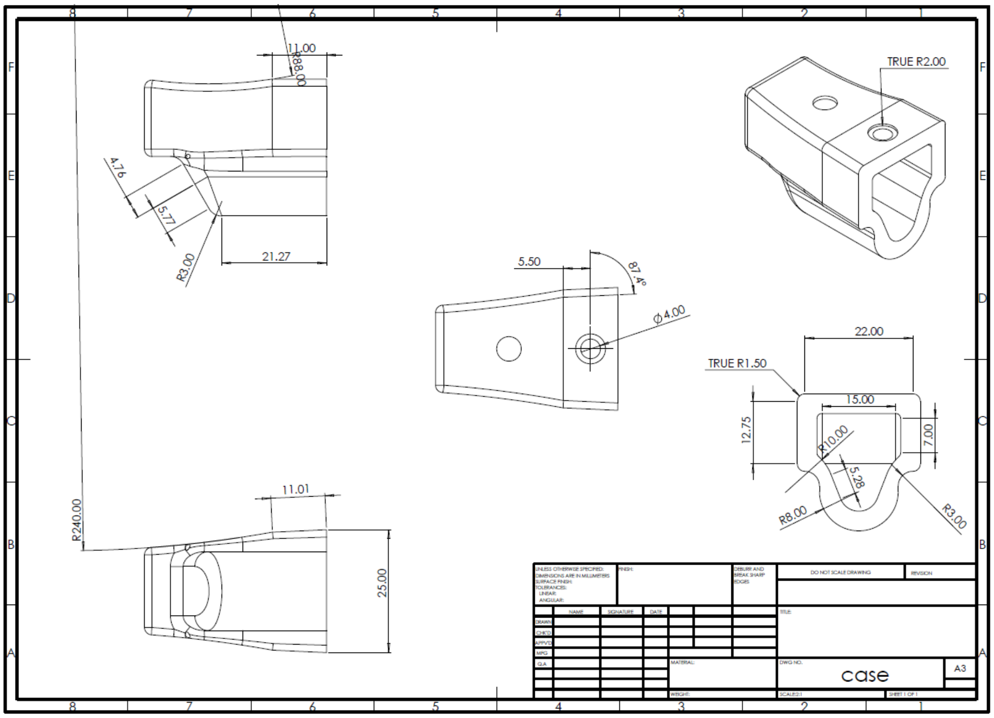
Version 3

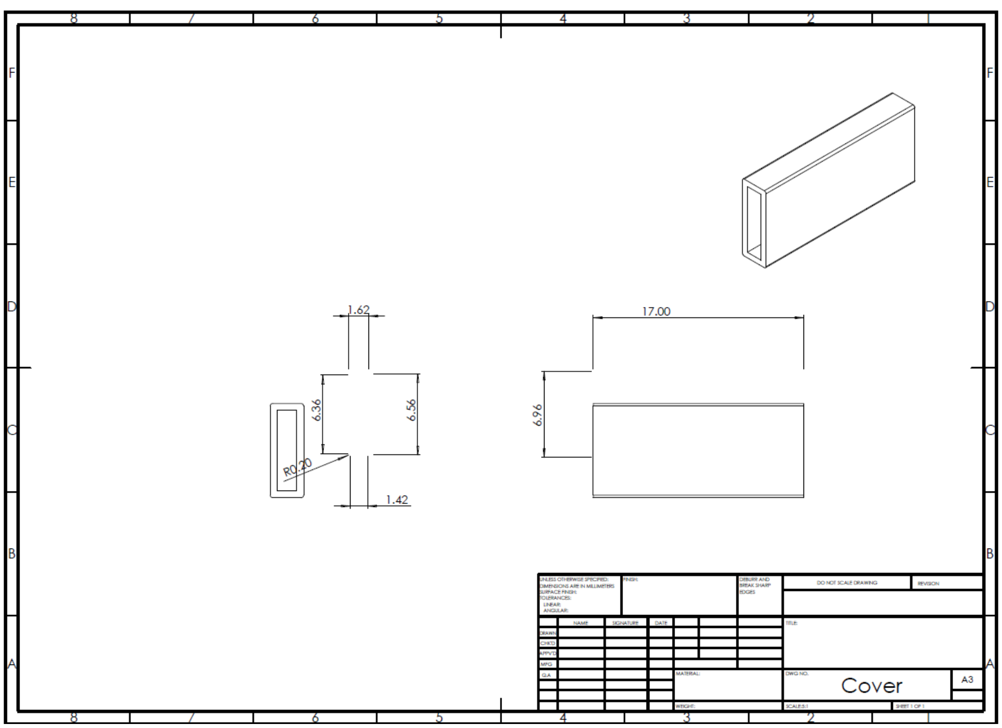


Ø3.00
Ø3.50
90°
4.76
17.26
1.42
6.26
Ø4.71
TRUE R2.00
4.01
1.85
16.76
7.00
R2.36
ac.plug.prongs
A3

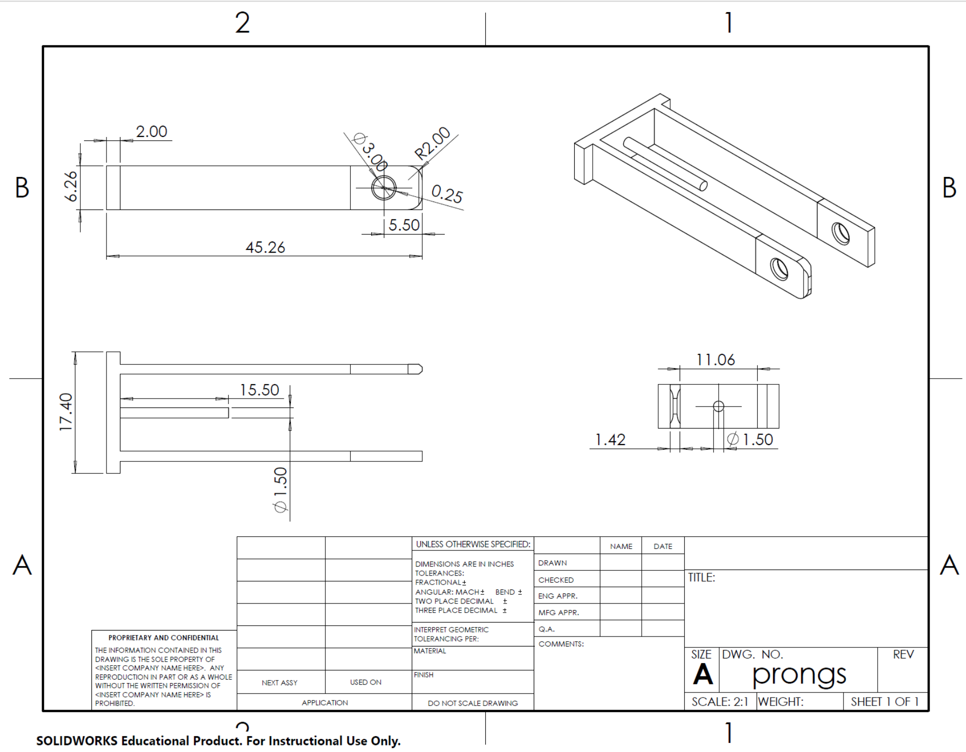


15.00
R1.00
2.02
0.20
7.00
6.86
68.53°
0.65
R10.00
Ø5.11
R4.00
90°
Ø4.00
10.00
acplug.inner
A3

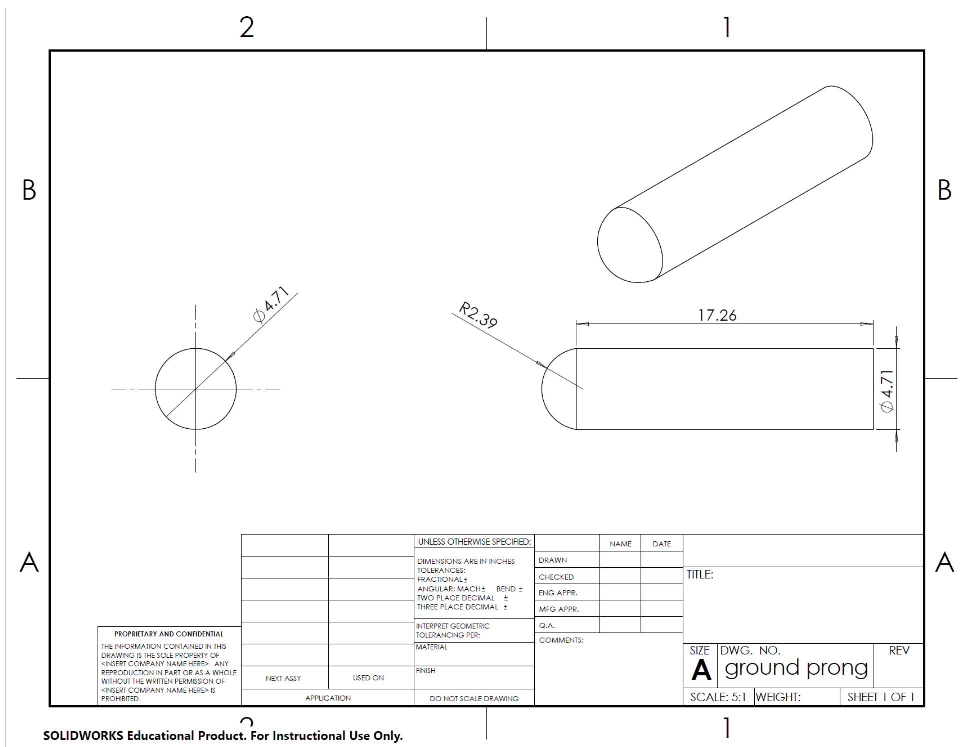

11.00
R88.00
4.76
5.77
R3.00
21.27
5.50
87.4°
Ø4.00
TRUE R2.00
22.00
TRUE R1.50
15.00
12.75
R10.00
7.00
5.28
R8.00
R3.00
11.01
R240.00
25.00
case
A3


1.62
17.00
6.36
6.56
6.96
R0.20
1.42
Cover
A3

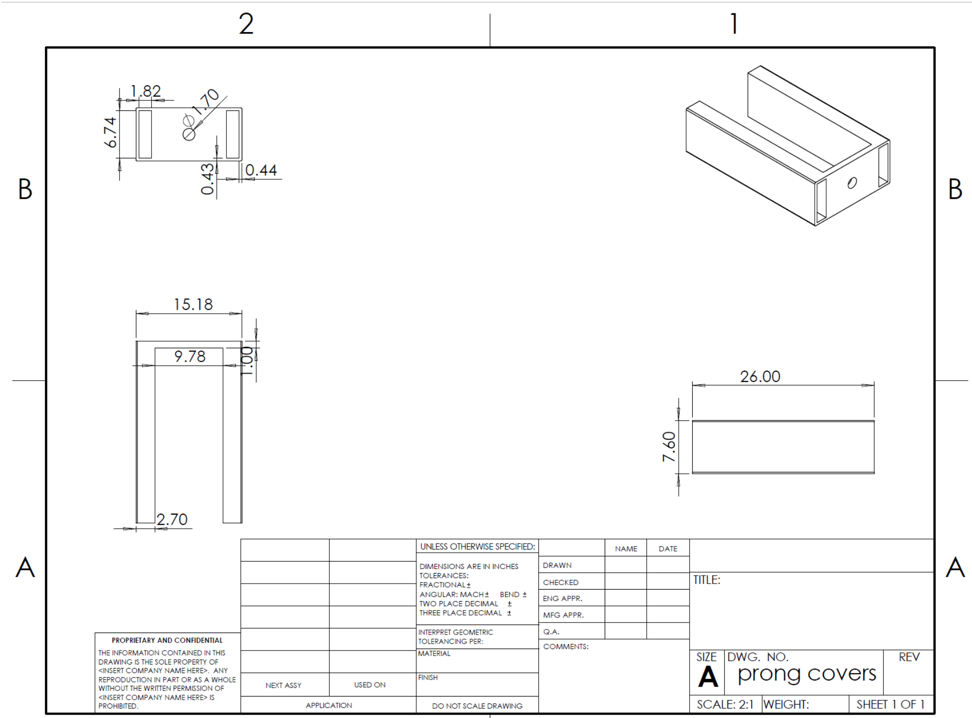
Version 4

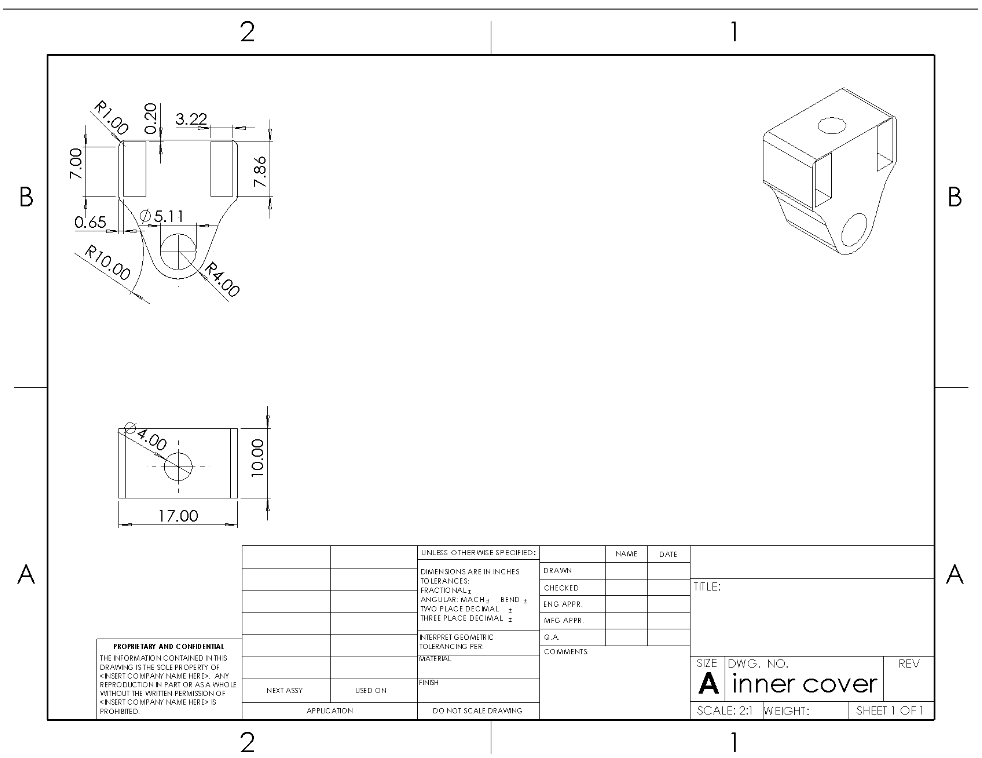


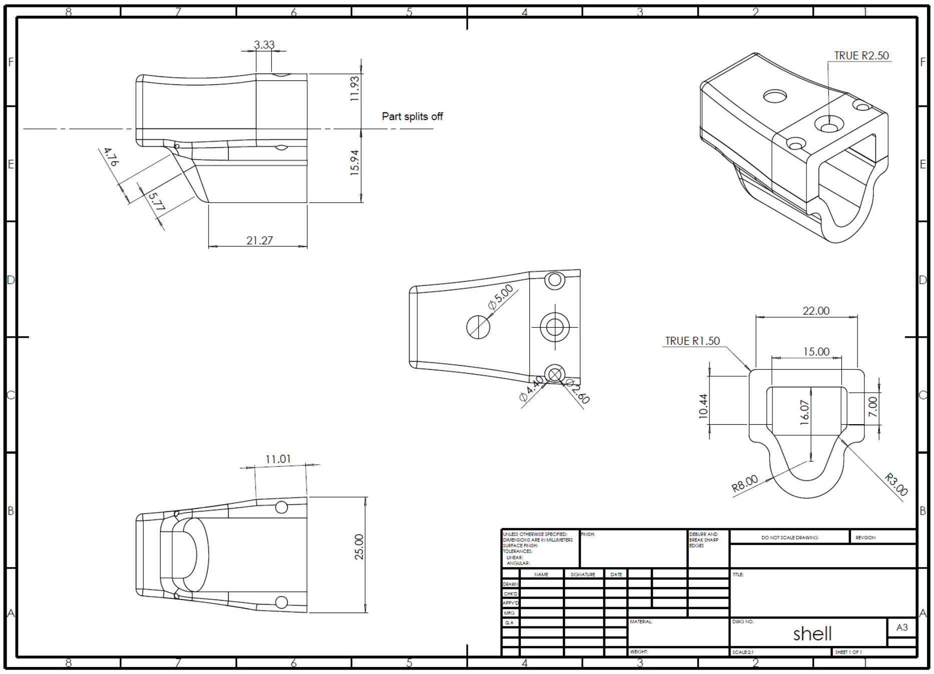
2.00
6.26
∅3.00
R2.00
0.25
5.50
45.26
15.50
17.40
∅1.50
11.06
1.42
∅1.50
SIZE
A
DWG. NO.
prongs
SCALE: 2:1
SHEET 1 OF 1
SOLIDWORKS Educational Product. For Instructional Use Only.



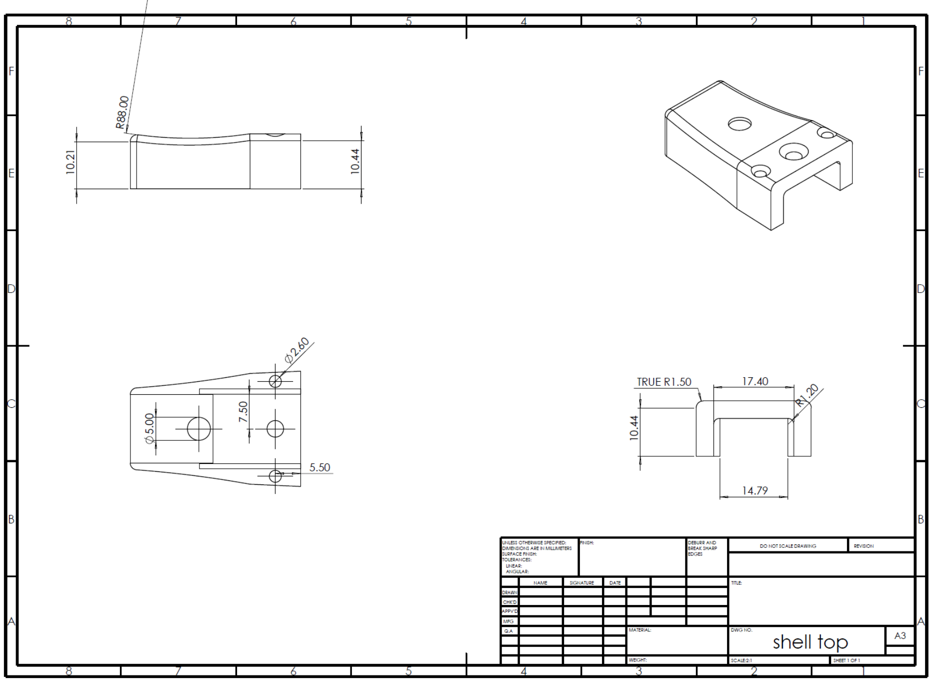
∅4.71
R2.39
17.26
∅4.71
SIZE
A
DWG. NO.
ground prong
SCALE: 5:1
SHEET 1 OF 1
SOLIDWORKS Educational Product. For Instructional Use Only.


2
1
B
A
1.82
⌀1.70
6.74
0.43
0.44
15.18
9.78
1.00
2.70
26.00
7.60
UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONAL±
ANGULAR: MACH± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL
FINISH
DO NOT SCALE DRAWING
NAME
DATE
DRAWN
CHECKED
ENG APPR.
MFG APPR.
Q.A.
COMMENTS:
TITLE:
SIZE
DWG. NO.
REV
A
prong covers
SCALE: 2:1
WEIGHT:
SHEET 1 OF 1
NEXT ASSY
USED ON
APPLICATION
PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

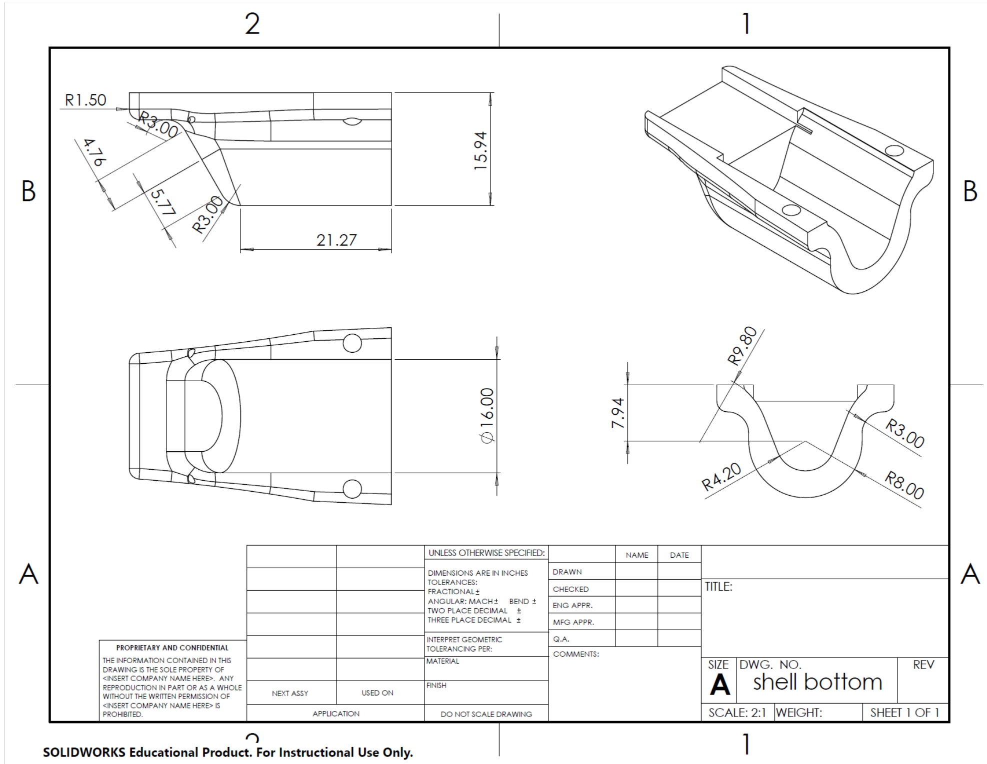

2
1
B
A
R1.00
0.20
3.22
7.00
7.86
0.65
⌀5.11
R10.00
R4.00
⌀4.00
10.00
17.00
UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONAL±
ANGULAR: MACH± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL
FINISH
DO NOT SCALE DRAWING
NAME
DATE
DRAWN
CHECKED
ENG APPR.
MFG APPR.
Q.A.
COMMENTS:
TITLE:
SIZE
DWG. NO.
REV
A
inner cover
SCALE: 2:1
WEIGHT:
SHEET 1 OF 1
NEXT ASSY
USED ON
APPLICATION
PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.



SOLIDWORKS Educational Product. For Instructional Use Only.



SOLIDWORKS Educational Product. For Instructional Use Only.


R1.50
R3.00
4.76
5.77
R3.00
21.27
15.94
∅16.00
7.94
R9.80
R3.00
R4.20
R8.00
UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONAL±
ANGULAR: MACH± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL
FINISH
DO NOT SCALE DRAWING
NAME
DATE
DRAWN
CHECKED
ENG APPR.
MFG APPR.
Q.A.
COMMENTS:
TITLE:
SIZE
A
DWG. NO.
shell bottom
REV
SCALE: 2:1
WEIGHT:
SHEET 1 OF 1
PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.
NEXT ASSY
USED ON
APPLICATION
SOLIDWORKS Educational Product. For Instructional Use Only.